Pico TNC User Manual

Introduction

Welcome to the **Pico TNC User Manual**. This document is your guide to setting up and using the Pico TNC — a modern re-implementation of a classic Packet Radio Terminal Node Controller (TNC).

A **TNC** is the device that makes packet radio possible. It sits between your radio and computer (or terminal) and takes care of the hard work: turning data into tones for transmission, decoding signals from the air, handling error correction, and following the AX.25 protocol used in amateur packet networks.

The Pico TNC is unique because it combines:

- **Modern hardware** — built on the inexpensive Raspberry Pi Pico microcontroller.
- **Classic software** — emulating the Z80 processor and command set of the popular TNC2 family.

This means the Pico TNC is both:

- **Beginner-friendly** — you can connect it to a radio, type a few basic commands, and be on the air with packet in minutes.
- **Historically compatible** — it supports the full TNC2 command set for running older packet software or experimenting with retro computing.

In this manual you'll find:

- Step-by-step instructions for flashing the firmware, wiring the TNC, and making your first connection.
- A quick guide to the handful of commands most hams use today.
- A complete reference to the original command set, for compatibility with legacy software or advanced use.

Whether you're new to packet radio, looking to explore APRS, or just curious about how earlier packet systems worked, the Pico TNC gives you a low-cost, open-source way to experiment and get on the air.

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What is Pico TNC?

A Terminal Node Controller (TNC) is a device that sits between your radio and your computer (or directly to your terminal) to send and receive packet radio data. It handles tasks like modulation & demodulation of signals, framing data into packets (AX.25), error detection, etc. The Pico TNC is an open-source Terminal Node Controller (TNC) designed for amateur packet radio applications. It is built around the Raspberry Pi Pico microcontroller and provides full TNC functionality, supporting the complete AX.25 Level 2 protocol suite. Unlike many modern TNCs that are limited to APRS (Automatic Packet Reporting System) operations, the Pico TNC emulates a classic HK-21 Pocket Packet TNC, making it compatible with legacy software and hardware.

What makes *Pico-TNC* special

- Built around the **Raspberry Pi Pico**, which provides modern microcontroller features + low cost.
- It emulates a **Z80 processor** running TNC2 type firmware (or behavior), meaning existing TNC2 commands & behavior are preserved.
- Supports **Bell 202 AFSK** modulation/demodulation in software (no separate modem chip required).
- Provides standard modes: *terminal mode* (using the built-in command set), *KISS mode* (for computer integration), AX.25 Level 2 compatibility.

Key benefits include:

- Low cost and ease of assembly using the readily available Raspberry Pi Pico.
- Support for both command-line interaction (emulating the HK-21) and KISS (Keep It Simple, Stupid) mode for integration with modern software like APRS digipeaters or chat applications.
- Experimental nature: This project is under active development and suitable for testing and hobbyist use.

What you'll need

- A Raspberry Pi Pico (board) loaded with the Pico-TNC firmware
- A radio capable of handling AFSK Bell 202 (commonly used in VHF packet)
- Required audio interfaces (audio in/out) or appropriate discriminator connection depending on radio
- Serial connection (USB or TTL) to a terminal or PC to send commands / receive data
- Power supply, cables, etc.

✅ **NOTE:** This manual is derived from the project's GitHub repository (https://github.com/pfiliberti/pico_tnc). For the latest updates, refer to the repository's README and documentation files.

Intended Audience

This manual is for amateur radio operators (hams) with basic electronics knowledge, familiar with packet radio concepts. Beginners should review introductory materials on AX.25 and TNCs.

Features

- **Full AX.25 Level 2 Support**: Handles framing, addressing, CRC checks, and retransmissions for unconnected and connected-mode operations.
- **Software AFSK Modem**: Encodes/decodes 1200 baud Bell 202 signals (mark: 1200 Hz, space: 2200 Hz) using the Pico's ADC/DAC.
- **Emulation of TNC2 type TNC**: Provides the same command set and behavior as the original 1980s hardware, including Personal Bulletin Board System (PBBS) functionality.
- **Interfaces**:
- USB serial for host computer connection (appears as a virtual COM port).
- 3.3V TTL serial for direct radio or legacy terminal connections.
- **KISS Mode**: Switchable for use with software that expects the KISS protocol (e.g., Xastir, Winlink).
- **Open Squelch Operation**: No hardware squelch required; relies on signal detection.
- **PBBS Support**: Built-in bulletin board for message storage and retrieval.
- Baud rate changes in the emulated TNC dynamically update the Pico's TTL serial port when used as a Console.

Limitations (Experimental Status):

- No native HF support (future: 1600/1800 Hz tones).

Hardware Requirements

- **Core Component**: Raspberry Pi Pico (original or Pico 2; RP2040-based).
- Low-pass filter for audio output (TX) to transceiver mic/audio in.
- Resistors/capacitor and protection diodes for audio input
- Audio in/out and PTT on TRRS jack (3.5mm or similar).
- PTT (Push-To-Talk) control if using VOX; GPIO for hardware PTT.
- **Serial**: USB Type-A/B for host, and or TRS Audio Jack for TTL serial.
- **Power**: 5V via USB..
- **Recommended:** Enclosure and a level shifter for RS-232 serial if needed.

Refer to the project schematic (available in the repository as `schematic.png`) for pinouts:

- GP0: TX audio out, GP26: RX audio in.
- Other GPIOs for PTT, TTL Serial Port and DIP switches.

✅ **NOTE:** Github repository located at: (https://github.com/pfiliberti/pico_tnc).
✅ **Assembly Tip**: Solder headers to the Pico and build a simple breakout board for audio/serial. Test audio levels to avoid overdriving the transceiver (typical: 20-50 mV pp for RX, 100-500 mV pp for

TX). Or use the pc board design in the github repo.

DIP Switch Settings

The Pico Tnc has a Dip Switch with 4 individual switches marked 1-4 and are in order from left to right. The rightmost switch 4 is used to enable/disable the leds. This can be used to save power when operating in the field on battery power as led's use quite a bit of power compared to the Pico's CPU. The remaining switches 1-3 form a combination that selects what Ports on the Pico Tnc are assigned to TNC Console or KISS according to the following combinations:

SW0	SW1	SW2	USB PORT	SERIAL	DESCRIPTION
OFF	OFF	OFF	CONSOLE	KISS	USB attached to TNC, Serial is KISS port
ON	OFF	OFF	KISS	CONSOLE	USB is KISS port, Serial attached to TNC
OFF	**ON**	OFF	CONSOLE	CONSOLE	Both USB and Serial are mirrored Console ports
ON	**ON**	OFF	KISS	KISS	Both USB and Serial are KISS ports
OFF	OFF	**ON**	CONSOLE	MESSAGE	Not yet implemented
ON	OFF	**ON**	MESSAGE	CONSOLE	Not yet implemented
OFF	**ON**	**ON**	N/A	N/A	Spare
ON	**ON**	**ON**	N/A	N/A	Pico Resets and Enters bootloader

✅ ***Tip:*** You do not have to manually reset the Pico Tnc when making a change to the dip switches. Once a change is made, it is automatically detected and the Pico Tnc will perform a reset and come up in the new mode.

✅ ***Tip:*** When using the Serial port for KISS mode the serial port parameters including speed are automatically set to 115200 baud,8 data bits ,no parity,1 stop bit. When using the Serial port for CONSOLE operation, the serial port parameters are set by the emulated TNC and can be changed using the **ABAUD**, **AWLEN** (data bits), and **PARITY** commands of the TNC. Setting the TNC **ABAUD** speed to 75 will be translated to 115,200 baud and setting it to 110 will be translated to 38,400. All other Baud rates remain the same 2400=2400, 9600=9600, etc. See **ABAUD** command for details.

✅ **Note:** When running with a Kiss Port be cautious how you configure your CALL-SSID for the software you are utilizing with the kiss port. For example if you already have assigned your call in the TNC using **MYCALL** and also assigned you call for the pbbs as CALL-1 using **MYMCALL** then you cannot use those calls again as they are being utilized by the TNC. You will need to us for example CALL-2 or CALL-3 just make sure they are not the same!

Status LEDs

The Pico Tnc has 4 Status Leds. They are labeled PTT, DCD(CDT), CON, and STA. You can locate these in the schematic diagram. On our designed PCB we ordered them so when looking at them they are ordered from right to left in the same order as described above. You can install any colored LED you desire. We chose Red for PTT, Green for DCD, Orange for CON and Blue for STA.

The Led's serve to indicate the following status of the Emulated TNC as follows:

- PTT – This LED is physically wired to the PTT line and lights up when the TNC is keying your radio and transmitting information.
- DCD – This LED lights up when the TNC has detected traffic on the channel. When lit it also prevents the PTT from the TNC until the channel is clear. Due to the nature of the open squelch code sometimes it flickers with noise on the channel and it may flicker at times when PTT is transmitting as well. This is not a problem and may be corrected in future firmware releases.
- CON – This LED lights up when your TNC is connected to another TNC. It shows the connection is established and active.
- STA – This LED lights up when the TNC has data in its buffers that it needs to transmit to another Station and will clear when the data has been transmitted and acknowledged by the other station.

Installation and Setup

Flashing the Firmware

Downloading the Firmware:

- Go to the github repository online @ https://github.com/pfiliberti/pico_tnc
- Scroll down and click **pico_tnc.uf2**
- Click **Download Raw File** and save to PC.

Preparing the Pico to receive the Firmware:

- Plug a USB A to USB Micro cable into the USB Micro port of the Pico.
- While pressing the **BOOTSEL button** on Pico, Plug the USB A cable end into the PC.
- Open Windows File Explorer and find the newly created **RPI-RP2 drive**, click to open.

✅ **NOTE:** There are 2 files created in the drive, INDEX.HTM and INFO_UF2.TXT. Disregard both.

Flashing the pico_tnc.uf2 Firmware:

- Find the pico_tnc.uf2 file downloaded previously.
- Drag the file into the RPI-RP2 drive.
- File will copy over and flash program the Pico.
- The open RPI-RP2 drive window will automatically close.

The Raspberry Pi Pico is now flashed. Configure the DIP SWITCHES to select Serial Console TNC access or KISS TNC functions, (See DIP_SWITCHES), and the TNCis ready for use.

Connecting to a Radio

1. Connect TX audio pin from RIG TRRS Jack to transceiver mic input.
2. Connect RX audio from transceiver discriminator to RIG TRRS Jack.
3. Wire PTT if needed to TRRS Jack.

4. Set transceiver to 1200 baud packet mode (e.g., FM, 145.010 MHz simplex for testing).
5. Connect host computer via USB serial (baud: 9600 default) using a terminal emulator (e.g., PuTTY, minicom).

✅ **Safety Note**: Use appropriate attenuation to prevent RF feedback. Test with dummy load first. Ensure compliance with local amateur radio regulations.

Operation

Power-On and Initial Connection

On the first power up, the TNC ram is fresh and so it does not know what baud rate or speed to connect to on the TTL Serial port. It is recommended to first use the USB Serial port to talk to it the first time. However if you do need to use the Serial interface, the default serial setting will be set to 9600 Baud, 7 Bits, Even Parity and 1 stop bit. If your forced to use another Baud rate like 2400 or 1200 on older equipment, the TNC will autobaud and probe for the correct speed. When you see the message: "**Please type a star(*) for auto-baud routine.**" press the * key on your keyboard to set the TNC to that speed.

Upon power-up, the TNC initializes the Z80 emulator and modem. Connect via USB Serial Port or via serial terminal at 9600 7E1. You'll see a prompt like:

```
Pico TNCEMU Emulated Z80 TNC Ver 0.60
|A
V-TNC
Virtual Machine TNC AX.25 Level 2 Version 2.0
Emulated Z80 Machine 32k Rom / RAM Mini PBBS ver 1.10 ase
Checksum $0E
cmd:
```

Type commands, followed by <CR>. Status indicators: LED on Pico blinks for activity; monitor serial for echoes.

Getting Started — First Use

Here's a typical workflow to get Pico-TNC working for the first time:

Step	What to do	What you should see or check
1	Connect the Pico-TNC to your computer via USB (or to a terminal)	Device appears; terminal opens, you see a prompt (if terminal mode)
2	Connect the radio: audio out from TNC to radio mic input,	Set levels low to start; check if

Step	What to do	What you should see or check
	audio in from radio speaker/discriminator to TNC's RIG jack	audio path works (e.g. tap-test)
3	Power up the system; make sure any LEDs or indicators light up	LED for power, possibly status LED
4	Enter basic command to identify firmware version, e.g. `VER` (or whatever the equivalent command is)	Response showing version & maybe firmware date
5	Configure the callsign and your address parameters (e.g. "MYCALL")	Confirm they're set using a query command
6	Try sending a test packet to another known station; listen for response	Packet comes through; check if decoded correctly
7	Switch to KISS mode if desired (command / switch) and test packet software on computer	Data flows in/out via KISS interface

✅ **NOTE:** TheTNC does not have a PERM command which is usually used to save the parameter changes in ram. Instead the TNC emulator detects a change to **TXDELAY**. It is recommended to set it to 50 but if it is already at 50 you can set it to 51 or 49 to save your parameters to the pico flash memory. If you care to you can do it again and set it back to 50 afterwards and it will save again. ***Again to save your TNC parmeters change TXDELAY.***

Hints

Modes of Operation

- **Command Mode (Default)**: Interactive TNC2 emulation for manual control and PBBS.
- **KISS Mode**: Enable with dip swich settings). In KISS, data is passed transparently without commands.

  To switch: Consult Dip Switch Settings.

Converse Mode vs Command Mode:

- In **Command Mode** you set parameters (`ABAUD`, `MYCALL`, etc.).
- In **Converse Mode** you type messages that are sent over packet.
- Switch between them with `CONVERSE` (or `K`) and `<CTRL-C>`.
- **Monitoring Traffic:**
- Use *`MONITOR ON`* to see other packets on the channel.
- Combine with *`MALL`* or *`BUDLIST`* for filtering.
- **Beaconing:**
- Beacon text (*`BTEXT`*) is what's sent; interval is set with *`BEACON`*.
- Use beacons sparingly on busy channels!

Sending and Receiving Packets

- **Converse Mode**: Use CONNECT callsign to establish a session.
- **Unconnected (UI) Frames**: Send raw data after *UNPROTO* dest via.
- **Monitoring**: Enable with *MONITOR ON* to see all heard packets.
- Incoming packets appear on serial; outgoing via keyboard input.

Example session:

A Brief History of Packet Radio and the TNC

Packet radio began in the late 1970s and early 1980s as a way to send digital data over amateur radio. Instead of voice, the radio transmitted structured packets of information, using the AX.25 protocol (a variant of X.25 used on early computer networks).

In those days:

- **Computers were slow**: Home and hobbyist computers often had limited memory and CPU power.
- **Dumb Terminals were used:** Sometimes instead of a full computer a teleprinter or dumb terminal where used such as the Dev VT100.
- **Serial ports were primitive**: Many terminals could only handle 300–1200 baud speeds reliably.
- **Hardware was simple**: Radios didn’t have built-in packet capability, so everything had to be handled by an external device.

The **Terminal Node Controller (TNC)** was invented to bridge this gap. It did the heavy lifting, framing packets, error checking, re-transmissions, so that even a very slow computer or dumb terminal could get on the packet network.

Because of that environment, TNCs developed a **large and detailed command set**.

- Many commands dealt with **flow control**, **parity bits**, or **baud rates** that matched the quirks of early 8-bit home computers.
- Others existed to let the TNC function **stand-alone** as a mailbox (PBBS), digipeater, or even a primitive network node without any computer attached.
- Different regions and repeater setups also needed specialized timing and compatibility tweaks.
- A lot of commands such as transparent mode allowed for software running on the computer to access the network without having to worry about the underlying network protocols.

Do we need all those commands today?

For most modern operators — **no**. Computers, radios, and packet software are much faster and more capable. Many of the fine-tuning parameters were critical in the 1980s but are rarely needed now unless:

- You're a **retro computing enthusiast** using an original terminal or 8-bit computer.
- You're running **legacy packet software** that expects a "real" TNC2 command set.
- You want to learn how things were done historically.

The Pico-TNC includes them because it **faithfully emulates the original TNC2 family firmware**. This means it is 100% compatible with old systems — but as a modern user, you can usually ignore most of them.

Common Commands Used Today

Here are the handful of commands you actually need in day-to-day packet operations:

- **MYCALL** — sets your station callsign. (Must be set before transmitting.)
- **MYMCALL** — sets your station's bbs callsign. (Must be set when bbs is active.)
- **MBOD** — Enables / Disables personal bbs mailbox.
- **MONITOR ON** — lets you watch packets on frequency.
- **CONNECT** — connect to another station or node.
- **DISCONNE (D)** — disconnect from a session.
- **CONVERSE (K)** — switch to chat mode once connected.
- **BTEXT / BEACON** — set a beacon message and control its timing.
- **UNPROTO** — set the destination path for unconnected packets (like APRS).
- **TXDELAY** — how long to wait before sending packet keying your transmitter's PTT line.
- **PERSIST** — threshold value for random number attempt to transmit.
- **SLOTTIME** — works with Persist command to achieve true p-persistent CSMA.

With just those commands, a new operator can do 90% of what packet radio offers today.

What is KISS Mode?

In the mid-1980s, a simpler approach was developed called **KISS (Keep It Simple, Stupid) mode**. Instead of typing TNC commands, the TNC passes raw packet frames back and forth with the computer using a minimal protocol.

- The computer (using software like **Xastir**, or an APRS client) handles the packet logic.
- The TNC just does the **modem work**: converting between radio audio tones and digital bits.
- KISS greatly simplifies software integration and avoids dealing with hundreds of TNC commands. However you must have software that is capable of handling the AX.25 Network Stack such as the AGWPE Engine.

Today, most packet applications use **KISS mode** because it makes the TNC behave like a simple modem peripheral rather than a standalone computer. But this also means you have to have a computer and all the software and complex configuration just to get on the air, whereas with the TNC all that is required is a serial connection!

The Pico Tnc software has been designed in such a way as to provide both support for **KISS** mode as well as standard serial console support and can do so simultaneously. With the correct DIP SWITCH settings either the USB Port or the TTL Serial Port can be set to be **KISS** or **CONSOLE** as well as both being **KISS** and both being **CONSOLE**. More options will follow with software updates.

The Personal BBS (PBBS)

In addition to handling packet connections and digipeating, the Pico TNC's emulated TNC includes a small **Personal Bulletin Board System (PBBS)**. This lets other stations connect directly to your TNC, leave messages, and read messages without requiring a separate computer.

The PBBS emulates what many TNC2-family devices offered in the 1980s, and while modest in size, it remains a fun and useful feature today.

Enabling the PBBS

The PBBS is disabled by default. To turn it on, set:

```
MBOD ON
MYMCALL <your-call-sign-SSID>
```

- **MBOD ON** — enables the mailbox system.

- **MYMCALL** — sets the mailbox callsign. Typically this is your call with an SSID, such as `N0CALL-1`, so your main call (`N0CALL`) can still be used for normal packet activity.

Once enabled, other stations can connect directly to the mailbox callsign and interact with the PBBS.

PBBS User Commands (when connected)

When another station connects to your PBBS callsign, they can use a small set of single-letter commands:

Command	Meaning	Example	Notes
L (LIST)	List messages	`L`	Shows message numbers, sender, recipient, and subject.
M (MINE)	List your messages	M	Shows message numbers, sender, recipient, and subject of messages addressed to your call.list
R n (READ)	Read message *n*	`R 3`	Reads message #3.
S (SEND)	Send a new message	`S N0CALL`	Prompts for subject and body. End with CTRL-Z.
K n (KILL)	Delete message *n*	`K 5`	Deletes message #5 (sysop privilege may apply).
H (HELP)	Display PBBS help	`H`	Summarizes available PBBS commands.
B (BYE)	Disconnect	`B`	Closes the session.

Sysop (Owner) Commands

From the TNC **command mode** (your local console), you can manage the PBBS directly:

Command	Purpose	Example
FILE	Show all stored messages	`FILE`
READ n	Read message *n*	`READ 2`
WRITE	Compose a new message	`WRITE` (then follow prompts)
KILL n	Delete message *n*	`KILL 4`

This lets you manage your mailbox without connecting over the air.

Notes and Limitations

- The PBBS has limited storage — keep messages short.
- Best practice is to dedicate a **secondary SSID** (e.g. `-1`) for the PBBS.
- You can disable it anytime with `MBOD OFF`.

Modern BBS Usage

While today’s packet networks usually rely on full-service BBS nodes or Internet-linked software, the PBBS remains handy for:

- Running a **simple personal mailbox** without a PC, useful in the field or in a grid down situatuon.
- Teaching newcomers how early packet systems worked.
- Retrocomputing and **demonstrating packet radio history**.

Command Reference

The Pico TNC supports the full TNC2 command set, as the Z80 code is a direct port. Commands are case-insensitive but traditionally uppercase. Syntax: COMMAND [PARAMETERS]<CR>.

This is a list of commands that the tnc firmware supports.

8BITCONV ON/OFF
Permits packet transmission of eight-bits data in the converse mode.

If 8BITCONV is OFF, the high-order bit (bit seven) of character received from the terminal is removed before the characters are transmitted in a packet. The standard ASCII character set requires only seven bits. The eight or final bit is used as a parity bit or ignored.

Command : 8BITCONV ON/OFF
Mnemonic: 8
Default : ON

Parameter : ON The high-order bit IS NOT stripped in Converse Mode.
OFF The high-order bit IS stripped in Converse Mode.

Refer to : AWLEN, PARITY, RXBLOCK

✅ **NOTE:** Setting bit seven in text characters transmitted over the air may cause confusion at the other end. If you need to transmit eight-bit data but don't want all the features of Transparnt Mode, set 8BITCONV ON and AWLEN 8. This may be desirable if you are using a special non-ASCII character set.

Since commands require only the standard seven-bit ASCII character set, bit seven is always removed in the Command Mode.

Set this command to ON if you use RXBLOCK.

AX25L2V2 ON/OFF

Command: AX25L2V2 ON/OFF
Mnemonic: A
Default: ON

Parameter : ON The TNc uses AX.25 level 2 version 2.0 protocol.
OFF The TNc uses AX.25 level 2 version l.0 protocol.

Some implementations of the earlier version of AX.25 protocol will not repeat version 2.0 AX.25 packets properly. This command provides compatibility with those older packet controllers.

ABAUD n (75 - 9600)

ABAUD displays the baud rate set by the auto baud routine when you first apply power to the unit or type RESET.

Command : ABAUD n (0 - 19200)
Mnemonic: AB
Default: 9600

Parameter: n Specifies the data rate of signaling speed in baud, on the serial I/O terminal port. The "n" should be **75***, **110***, 300, 600, 1200, 2400, 4800, and 9600. **Anything above 9600 is set to 9600.**

Refer to: RESTART, RESET, RAMTEST

Use ABAUD to specify a terminal baud rate that will become active on the next power up or RESTART. A warning message reminds you of this.

If you plan to change terminals but want to retain all of the RAM parameter values, set ABAUD, AWLEN, and PARITY to the new terminal's characteristics while you are still connected to the old terminal. Then turn off the TNC, change to the terminal and turn the TNC on again.

✅ ***NOTE:** Speeds of 75 and 110 are special. They are translated by the tnc emulation to higher baud rates that were not normally achievable in the tnc software. 75 will set the port to 115,200 baud and 110 to 38,400 baud.

AUTOLF ON/OFF

Command: AUTOLF ON/OFF
Mnemonic: AU
Default: ON

Parameter: ON A line feed character <LF> is sent to the terminal after each carriage return character <CR>.

OFF A <LF> is NOT sent to the terminal after each <CR>.

Refer to: LFADD, RXBLOCK

AUTOLF is included to maintain compatibility with other packet radio controllers. If your display terminal does not line-feed during packet reception, check this command.

AUTOLF affects only terminal displays and it does not affect sending packet data.

If the other station you are talking to reports overwriting of packets from your station, use LFADD command.

To use **RXBLOCK** command, set **AUTOLF** OFF to accurately display the data block length.

AWLEN n

Command: AWLEN n
Mnemonic: AW
Default: 7

Parameter: n 7 or 8 specifies the number of data bits per word.

Refer to: 8BITCONV, PARITY, RESTART

The parameter value defines the digital word length used by the serial input/output (IO) terminal port and your computer or terminal program. The new value affects only after RESTART or power On/Off reset.

✅ **NOTE:** Use default value (7) for most packet operations, such as conversation, working electronic mail and bulletin board systems and transmission of ASCII files.

If eight-bit words are sent to the TNC in the Command or the Converse modes, the eighth bit is normally removed, leaving a standard ASCII character, regardless of the setting of AWLEN.

The only time all eight data bits of each character must be retained is to send executable files or other special data.

✅ **NOTE:** Set AWLEN to 8 and use the Transparent Mode.

You can also use Converse Mode and set AWLEN 8 and 8BITCONV ON. However, you must precede the Converse Mode special characters with the PASS character in the data you send.

✅ **NOTE:** Set AWLEN to 8 if you use RXBLOCK command.

AXDELAY n

Command: AXDELAY n
Mnemonic: AXD
Default: 0

Parameter: n 0 - 120 specifies a key-up delay for voice repeater 10mS intervals.

Refer to: AXHANG, TXDELAY, DWAIT

AXDELAY acts in conjunction with AXHANG. It specifies the period of time the controller will wait, in addition to the normal delay set by TXDELAY, after keying the transmitter and before data is sent.

Packet groups using a standard "voice" repeater to extend the range of the local network may need to use this feature. Repeaters with slow electromechanical relays, split sites, auxiliary links, or other circuits which delay transmission for some time after the RF carrier is present, require some amount of time to get the RF on the air. If you are using a repeater that has not previously been used for packet operations, try various values to find thebest value. If other stations have been using the repeater, check with them for the proper setting.

AXHANG n

Command: AXHANG n
Mnemonic: AXH
Default : 0

Parameter: n 0 - 250 specifies voice repeater hang time in 100mS intervals.

Refer to: AXDELAY, DWAIT

A numeric value can be used to increase channel efficiency when an audio repeater has a hang time greater than 100milliseconds. For a repeater with a long hang time, it is not necessary to wait for the repeater key-up delay after keying he transmitter if the repeater is still transmitting.

When the TNC has heard a packet sent within the hang period, it does not add the repeater key-up delay (**AXDELAY**) to the key-up time. The key-up time is determined by **TXDELAY** + **DWAIT**. Therefore, set **AXHANG** should be set less than repeater hang time.

If you are using a repeater that hasn't been used for packet operation before, try various values to find the best value. If other packet stations have been using the repeater, ask them for the appropriate setting.

BEACON EVERY/AFTER n

Command: BEACON EVERY/AFTER n
Mnemonic: B
Default: EVERY 0

Parameter: EVERY Send the beacon at regular intervals.
AFTER Send the beacon once after the specified time interval without packet activity.

n 0-250 specifies timing in 10 Sec.intervals. 0 Zero disables the beacon function.

Refer to: BTEXT, UNPROTO

The BEACON command sets the conditions under which your packet beacon will be transmitted.

NOTE: A beacon frame contains the text you have typed into the **BTEXT** message in a packet addressed 'CQ' or other **UNPROTO** address.

NOTE: A beacon frame may be sent directly, and also sent via the digipeat addresses specified by the **UNPROTO** command.

When the keyword EVERY is specified, a beacon packet is sent every n times 10 seconds. This mode can be used to transmit packets for testing purposes.

When AFTER is specified, a beacon is sent after n times 10 seconds have passed without packet activity.

NOTE: The beacon is sent only once until further activity is detected.

This mode can be used to send announcements or test messages only when packet stations are on the air.

Tip: Proper choice of 'n' avoids cluttering a busy channel with lots of unnecessary transmissions.

BKONDEL ON/OFF

Command: BKONDEL ON/OFF
Mnemonic: BK
Default: ON

Parameter: ON The sequence <BACKSPACE> <SPACE> <BACKSPACE> is echoed when a character is deleted from the input line.

OFF The <BACKSLASH> character <\> is echoed when a character is deleted.

Refer to: **REDISPLA**

BKONDEL determines how character deletion is displayed in Command or Converse mode.

The <BACKSPACE><SPACE><BACKSPACE> sequence updates the video display screen.

NOTE: Set **BKONDEL** ON if you are using a video display terminal or computer.

On a printing terminal the <BACKSPACE> <SPACE> <BACKSPACE> sequence will result in over-typed text.

NOTE: Set **BKONDEL** OFF if you have a paper output display, or if your terminal does not respond to the <BACKSPACE> character <CNTRL-H>. The TNC displays a <BACKSLASH> for each character you delete. You can see a display of the corrected input by typing the redisplay-line character set by the command **REDISPLA**.

BTEXT text

Command: BTEXT text
Mnemonic: BT
Default: Empty

Parameter: Any combination of characters and spaces, up to a maximum length of 239 characters.

Refer to: **BEACON, PASS**

BTEXT is the content of the data portion of a beacon packet. The default text is an empty string (no message).

NOTE: BEACON packets are discussed in more detail under the **BEACON** command.

You can send multiple-line messages in your beacon by including <CR> characters in the text. <CR> is inserted by typing the **PASS** character (CTRL-V) before the <CR>.

NOTE: The pass character is set by the **PASS** command.

If you enter a text string longer than 239 characters, the command line is ignored and the following error message appears.

?too long

NOTE: Use a "%" or "&" as the first characters in the beacon text to clear the BTEXT text without issuing the RESET command.

Although the subject of beacon is conventional in packet circles, the following four suggestions will help you use the feature intelligently and benefit the packet community:

- Do not type your call sign in BTEXT. The normal packet header shows it for you.
- Do not fill your BTEXT with screen graphics such as asterisk, parentheses, colons and semicolons, etc. Use BTEXT for some significant information.
- Do not use BTEXT to tell the world that your "DIGIPEATER IS ON". Put this information in your CTEXT message so that it is seen by any station that connects you, the only one who can make use of the information.
- After you have beaconed for a week or two and the packet community has learned who and where you are, follow the practice used by more experienced packeteers. **SET BEACON EVERY 0!!!**

BUDLIST ON/OFF

Command: BUDLIST ON/OFF
Mnemonic: BU
Default: OFF

Parameter: ON Ignore frames from station which are not in the **LCALLS** list.
OFF Ignore frame from stations which are in the **LCALLS** list.

Refer to: LCALLS, MONITOR

BUDLIST works together with the **LCALLS** command, which allows you to set up a call sign list. These commands determine which packets are displayed when you set MONITOR to ON. **BUDLIST** allows you to specify whether the call signs in the list are those you wish to ignore, or if they are the ones you want to displayed.

If you want your TNC to display packets only from a limited list, use **LCALLS** to enter the list of call signs, and then set **BUDLIST** to ON. This feature enables you to have the TNC watch for a particular station while you converse with another station.

If you want your TNC to ignore packets from a limited list, use **LCALLS** to enter the list of call signs, and then set **BUDLIST** to OFF. This is handy when you want to ignore a bulletin board on the frequency, for example, while you monitor other conversations.

CONNECT call1[VIA call2[,call3...,call9]]

Command: CONNECT call1[VIA call2[,call3...,call9]]
Mnemonic: C

Default: Immediate command

Parameter: call1 Call sign of the distant station to which you wish to be connected.
call2 Optional call sign(s) of up to eight digipeaters via which you want to be repeated to reach the other station.

Refer to: DISCONNE

CONNECT sends a connect request to station 'call,' directly or via one or more digipeaters. Each call sign can include an optional SSID "n", immediately after the call sign.

The part of the command line shown in brackets below is optional. The double-bracketed text, 'call3...,call9' is also optional and is used only when 'VIA call2' is used, that is, when you are connecting through one or more digipeaters.
✅ **NOTE:** (The brackets and quotation marks are used in this text only for clarity. Do not type them!)

VIA call2[,call3...,call9]

✅ **NOTE:** Type the digipeater fields in the exact sequence you wish to use to route your packets to the destination station "call."

If you type **CONNECT** while your TNC is already connected, or trying to connect to or disconnect from another station, your monitor will display.

Link state is:CONNECT in progress

If the other station does not "ack" your connect request after the number of tries specified by **RETRY**, the **CONNECT** command is canceled and your monitor displays:

*cmd: *** Retry count exceeded*
**** DISCONNECTED: (call sign)*

To connect directly to KF7PSM, you would type:

`CONNECT KF7PSM` (or C KF7PSM)

To connect to W6YEY using W6UOU (with whom you can easily connect) and W6WNE (who is near W6YEY) as digipeaters, type:

`CONNECT W6YEY VIA W6UOU, W6WNE`

Type CONNECT or 'C' without arguments to see the link status and the number of unacknowledged, outstanding packets.

CANLINE n

Command: CANLINE n
Mnemonic: CAN
Default: $18 <CTRL-X>

Parameter: n 0 to $7F (o to 127 decimal) specifies an ASCII character code.

Refer to: CANPAC

CANLINE changes the CANCEL-LINE input editing command character.

The parameter "n" is the ASCII code for the character you want to use to cancel an input line. The following example show how to enter the code in hex and decimal.

CANLINE $15 (hex)
CANLINE 21 (decimal)

Either method sets the cancel-line character to <CTRL-U>.

When you use the CANLINE character to cancel an input line in the Command Mode, the line is terminated with a<BACKSLASH> character and new prompt (cmd:) appears.

When you cancel a line in the Converse Mode, only the <BACKSLASH> and a new line appear.

TIPS:

1. You can cancel only the line you are currently typing.
2. Once you type a <CR>, you cannot use the cancel-line character to cancel an input line.
3. Use CANPAC character to cancel the entire packet.
4. If your send-packet character is not <CR>, the cancel-line character cancels only the last line of a multi-line packet.

NOTE: Like all other input editing features, line cancellation is disable in the Transparent Mode.

CANPAC n

Command: CANPAC n
Mnemonic: CANP
Default: $19 <CTRL-Y>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII character code.

Refer to: CANLINE

CANPAC changes the CANCEL-PACKET input editing command character.

The parameter "n" is the ASCII code for the character you want to type in order to cancel an input packet.

✅ **NOTE:** You can enter the code in either hex or decimal.

When you cancel a packet in the Converse Mode, the line is terminated with a <BACK SLASH> character and a new line.

✅ **NOTE:** You can only cancel the packet that you care currently entering.

Once you have typed the send-packet character, or waited **PACTIME** (if **PACTIME** is enabled), the packet cannot be canceled even if it has not been transmitted.

Like other input editing features, packet cancellation is disabled in the Transparent Mode.

The CANCEL-PACKET character also cancels the display output in the Command Mode. If the TNC is in the Command Mode and you type the CANCEL-PACKET character, any characters that would be typed on the screen (except those choed) are"discarded" by the TNC.

✅ **TIPS:**

1. Typing the cancel-output character a second time restores normal output.
2. To see how this works, type DISPLAY followed by a <CTRL-Y>.

The command list display will stop and you will not see any response from the TNC.

To verify that the display is back to normal, type <CTRL-Y>, and then type DISPLAY again.

Use the CANCEL-DISPLAY feature if you inadvertently do something that cause the TNC to generate large amounts of output to the terminal, such as entering the DISPLAY command or setting **TRACE** ON.

✅ **NOTE:** If the TNC is in the Converse or Transparent Mode and you want to cancel display output, you must exit to the Command Mode before you type the CANCEL-PACKET character.

CBELL ON/OFF

Command:	CBELL ON/OFF
Mnemonic:	CB
Default:	OFF

Parameter:	ON Three BELL characters <CTRL-G> ($07) are sent to your terminal with the '*** CONNECTED to (call sign)'message. OFF BELLS are not sent with the CONNECTED message.

NOTES:

1. Set CBELL ON if you want to be notified whenever someone connects to your station.
2. If CBELL is on and MFILTER contains the character ($07), you can be sure that whenever your terminal beeps there is a connection for you. At no other time will you hear a beep while the TNC is in the Packet Mode.
3. This feature works only the stream is set to "K" when someone connected to **MYMCALL**.

CHECK n

Command: CHECK n
Mnemonic: CH
Default: 30

Parameter:: n 0 to 250 specifies the check time in 10 seconds intervals.
0 Zero disables this feature.

Refer to: AX25L2V2, TOUT
CHECK sets a time-out value for a packet connection, and depends on the setting of **AX25L2V2**.

Without the CHECK feature, if your TNC was linked or 'connected' to another station and the other station seemed to 'disappear', your TNC would remain in the connected state indefinitely, refusing connections from other stations.

This might happen if propagation changes unexpectedly or an intermediate digipeater station fails or is shut down while you and theother station are connected 'via' that digipeater.

Your TNC tries to prevent this sort of 'lockup' from occurring by sending a new connect request packet when specified time elapses without any packets being received from the other TNC.

If Perversion 2 link is inactive for CHECK times ten seconds, your TNC tries to save the link by starting a reconnect sequence. The TNC enters the 'connect in progress' state and sends SABM (connect request) frames. In addition, the TNC adds a random time of up to 30 seconds each time CHECK is used.

✅ **NOTE:** If AX25L2V2 is ON and packets have not been received from the other station for 'n' times 10 seconds, your TNC sends a 'check packet' to test if the link still exists to the other station.

The 'check' packet frame contains no information, but is interpreted by the other station's TNC to see if it is still connected. If the other station's TNC is still connected, it sends an appropriate response packet.

CLKADJ n ***(REDACTED!)***

Command: CLKADJ n
Mnemonic: n
Default: 0

Parameter: n 0 to 65535 specifies a correction factor that is applied to real time clock

Refer to : **DAYTIME**

CLKADJ provided you with a limited ability to correct the TNC real time clock. A value of 0 (zero) causes no corretion to be applied. If the value is anything else the corection factor was calculated as: relative clock speed in % = 100-(9.16667 x 1/n)

✅ **NOTE:** This is no longer used as the emulator uses the pico's internal real time clock which is more accurate.

CMDTIME n

Command: CMDTIME n
Mnemonic: CM
Default : 1

Parameter: n 0 to 250 specifies time-out value in 1 second intervals while in Transparent Mode.

Refer to: COMMAND, TRANS

If 'n' is 0 (zero) you will have to send the BREAK signal or interrupt power to the TNC to exit from Transparent Mode.

CMDTIME sets the time-out value in Transparent Mode. A guard time of 'n' seconds allows escape to the Command Mode from the Transparent Mode, while permitting any character to be sent as data.

The same Command Mode entry character (default <CTRL-C>) that is used to exit from the Converse Mode is also used to exit the Transparent Mode, although the procedure is different.

NOTES:

1. The Command Mode entry character is set by COMMAND.
2. Three Command Mode entry character must be entered less than 'n' seconds apart, with no intervening characters, after a delay of 'n' seconds following the last characters you typed.
3. After a final delay of 'n' seconds, the TNC exits the Transparent Mode and enters the

Command Mode.
4. You will then see the normal Command Mode prompt: *cmd:*

The following diagram illustrates this timing:

```
last      first     second    third          TNC
terminal   command   command   command      now in
input     mode      mode      mode        command
|        entry     entry     entry          mode
|        character  character  character      |
|          |          |         |        |
|          |          |         |        |
|          |          |         |        |
|          |          |         |        |
|<--- longer--->|<---shorter-->|<--shorter-->|< -- n --->|
 than 'n'      than 'n'     than 'n'
```

CMSG ON/OFF

Command: CMSG ON/OFF
Mnemonic: CMS
Default: OFF

Parameter: ON The recorded CTEXT message is sent as the first packet after a connection is established by a connect request from another station.
OFF The text message is not sent at all.

Refer to: CTEXT, CMSGDISC

CMSG enables or disables automatic transmission of the **CTEXT** message when your TNC accepts a connect request from another station.

NOTES:

1) Set CMSG ON to tell callers that you are not available to answer calls manually when they connect to your TNC.

2) Set CMSG OFF when available to operate or answer calls manually.

CMSGDISC ON/OFF

Command: CMSGDISC ON/OFF
Mnemonic: CMSGD
Default: OFF

Parameter: ON Cause an automatic disconnect upon completion of CTEXT message.
OFF The TNC will remain connected after it sends the CTEXT message.

Refer to: CTEXT, CMSG

Normally, you will want to set CMSGDISC to ON. If CMSGDISC is set to OFF and another station connects to you, your station will remain in a connected state until either you or the other station issue a **DISCONNECT**.

This feature helps prevent your station from being hung up by a distant station that connects you, but for some reason cannot disconnect, such as fading band conditions. If both CMSG and CMSGDISC are set to ON, link is disconnected after sending out CTEXT message. If no messages in CTEXT the link is also disconnected as normal disconnect. A normal disconnect process is performed regardless the setting of CMSG.

NOTE: If **CMSGDISC** is set to ON, make sure that the **CMSG** is also ON. If the **CMSG** is set to OFF, the link will be disconnected after your TNC sends out the first information frame.

NOTE: CMSGDISC needs to be set to OFF before using CONNECT to try and connect to another station. This is a bug in the TNC firmware that if you leave it ON will cause an immediate disconnect from the remote station.

COMMAND n

Command: COMMAND n
Mnemonic: COM
Default: $03 <CTRL-C>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII character code.

Refer to: CMDTIME, CONVERSE, TRANS

COMMAND is used to change the Command Mode entry character. You can enter the code in either hex or decimal. Type the COMMAND character to enter the Command Mode from Converse Mode.

You will not see a response if you type the Command Mode entry character while the TNC is already in the Command Mode.

To enter the Converse Mode type: CONVERSE

Now, all characters typed on the keyboard and characters sent from a disk or tape files are transmitted as packet data.

To return to the Command Mode, type the COMMAND character usually the default <CTRL-C>. The Command Mode prompt appears to indicate a successful exit to the Command Mode. The display might look like this:

```
cmd: CONVERSE
Hello World! I'm on the air on packet radio! [type <CTRL-C>]
cmd:
```

Refer to the CMDTIME and TRANS to escape from the Transparent Mode.

NOTE: Some operating systems such as Linux use <CTRL-C> to break out of a program. The use of CTRL-C in this case will not work for the COMMAND chaacter. You can change it to something else like <CTRL-E> by typing the following:

```
COMMAND $05
```

CONMODE CONVERSE/TRANS

Command:	CONMODE CONVERSE/TRANS
Mnemonic:	CONM
Default:	CONVERSE
Parameter:	CONVERSE Your TNC automatically enters the Converse Mode when a connection is established.
	TRANS Your TNC automatically enters the Transparent Mode when a connection is established.
Refer to:	**CONVERSE, TRANS**

CONMODE selects the mode your TNC uses after entering the CONNECTED state.

The connection may result either from a connect request received from another station, or a connection initiated by a CONNECT command that you have typed.

NOTES:

1. Set CONMODE to CONVERSE for most packet operations.
2. Set CONMODE to TRANS if you are using the Transparent Mode for a bulletin board program, so that the correct mode will be entered when your bulletin board receives a connect request.
3. If the TNC is already in Converse or Transparent Mode when the connection is completed, the mode will not be changed.

CONOK ON/OFF

Command: CONOK ON/OFF
Mnemonic: CONO
Default: ON

Parameter: ON Connect requests from other stations are accepted.
OFF Connect requests from other stations are not accepted.

Refer to: BBSMSGS, USERS

The **CONOK** command determines the action that the TNC will take when it receives a connect request from another stations.If **CONOK** is set to ON, the request is acknowledged, the normal connect message is displayed, and the TNC will enter either the Converse or Transparent Mode (depending upon how you have **CONMODE** set.)

If **CONOK** is set to OFF and the TNC is not in the Transparent Mode, it will display a connect request message:

Connect request: <call sign>

The TNC also issue a DM packet, like a "busy signal". to the requesting station. If you wish, you can then issue your own connect command. When your TNC receives a DM packet in response to a connect request, it will display:

**** <call sign> station busy*

If you are connected with several other stations (connected same number of stations set by USERS), your TNC will respond to the same as above.

If, for example, you wish to leave your station on as a digipeater, you could set CONOK to OFF until you are ready to communicate with another station. When another station attempts to connect to your TNC, it will be able to see that your station is not available for a connection, but can still be used as a digipeater.

CONPERM ON/OFF

Command: CONPERM ON/OFF
Mnemonic: CONP
Default : OFF

Parameter: ON The current connection on the current channels are not allowed to enter the disconnected state.
OFF The current channel can be connected to and disconnected from other stations.

Refer to: RELINK, RETRY, TRIES

When it is ON, **CONPERM** forces the TNC to maintain the current connection, even when frames to the other station exceed RETRY attempts for an acknowledgment. RESTART and power off/on cycling do not affect this connected state.

If disconnection is necessary, set this command to OFF and proceed the normal disconnect sequence.

CONPERM works only when a connection is established. It functions on a channel-by-channel basis when multiple connections are allowed.

CONPERM allows connections on other channels to operate normally. For example, automatic disconnect based on RETRY,when used under conditions such as:

- Certain networking applications.
- Meteor scatter.
- Other noisy, less reliable links.

CONPERM ON may be advantageous when you use full-duplex continuous mail forwarding or traffic links.

CONSTAMP ON/OFF

Command: CONSTAMP ON/OFF
Mnemonic: CONS
Default: OFF

Parameter: ON Connect status messages are time stamped.
OFF Connect status messages are not time stamped.

Refer to: DAYTIME, DAYUSA

CONSTAMP activates time stamping of *** CONNECTED status messages.

If **CONSTAMP** is ON and **DAYTIME** (the TNC's internal clock) is set, date and time information generated in the TNC is available for bulletin-board programs or other host computer applications.

Date and time must be set initially by the DAYTIME command before time stamping will occur. For example, if **CONSTAMP** is ON and the date and time have been set in the TNC, a connect and disconnect sequence appears as follows:

```
cmd: connect W6YEY
cmd: 10:55:23:26 *** CONNECTED to W6YEY
cmd: disconnect
cmd: 10:55:59:26 *** DISCONNECTED: W6YEY
```

The CONNECT and DISCONNECT command can be abbreviated as shown below. The results are identical.

```
cmd: c W6YEY
10:56:22:18 *** CONNECTED to W6YEY
cmd: d
10:56:32:29 *** DISCONNECTED: W6YEY
```

CONVERSE

Command:	CONVERSE
Mnemonic:	CONV or K
Default:	Immediate Command

CONVERSE is an immediate command that causes the TNC to change form the Command Mode to the Converse Mode.

When the TNC is in the Converse Mode, all characters typed from the keyboard or sent from a disk file are processed and transmitted by your radio.

NOTE: To return the TNC to the Command Mode from the Converse Mode, type the Command Mode entry character (default is <CTRL-C>).

CPACTIME ON/OFF

Command:	CPACTIME ON/OFF
Mnemonic:	CP
Default:	OFF

Parameter: ON Packet transmit timer is used in the Converse Mode.
OFF Packet transmit timer is not used in the Converse Mode.

Refer to: PACTIME

CPACTIME activates automatic, periodic transmission of packets while the TNC is in the Converse Mode. It may be used for several types of computer communications bulletin board or host computer operation when you do not want the full Transparent Mode features.

NOTES:

1. When **CPACTIME** is ON, characters are packetized and transmitted periodically like they are in the Transparent Mode. Local keyboard editing and display features of the Converse Mode are available; software flow control can be used.
2. Refer to the **PACTIME** command, (which controls the rate and mode of packet assembly) for a discussion of how periodic packetizing works.
3. Set a CR OFF in this mode. When a CR is ON, the send-packet character is inserted in the data being packetized even though it was not typed.
4. To include <CR> characters in transmitted packets, set SENDPAC to a character not normally used (e.g., <CTRL-P>); the TNC then treats a <CR> as an ordinary character.

Setting **CPACTIME** ON transmits your text soon after you type it, in short bursts of a few characters. The other station may break in at will. Some operators find it easier to chat in this mode; long delays are eliminated while long packets are being typed.

CR ON/OFF

Command: CR ON/OFF
Mnemonic: CR
Default: ON

Parameter: ON The send-packet character, normally <CR>, is added to all packets sent in the Converse Mode.

OFF The send-packet characters is not added to packets.

Refer to: SENDPAC

When CR is set to ON, all packets that are sent in the Converse Mode will include, as the last character of the packet, the send-packet-character. This character causes the packet to be sent. If **CR** is set to OFF, the send-packet character is interpreted only as a command to the TNC, and is not included in the packet. In addition, it will not be echoed to the terminal.

Set **CR** ON and **SENDPAC** to $0D for a natural conversation mode. each line is sent whenever you enter a <CR>, and arrives at the other station with a <CR> at the end of the line.

NOTE: If the other station reports overprinting of the lines on his display, you can set **LFADD** to ON or the other station can set **AUTOLF** to ON.

CSTATUS

Command: CSTATUS
Mnemonic: CS
Default: Immediate command
Refer to: CONPERM

CSTATUS is an immediate command that is used during multiple connections. When you type CSTATUS, your monitor displays:

- The number of each logical channel.
- The link state of all ten logical channels.
- The current input/output channel - the one you are using.
- Whether each channel connection is 'permanent.' (Refer to CONPERM)

Depending upon your use of multiple connections and **USERS** command, **CSTATUS** will display the following.

```
NOT CONNECTED TO ANY STATION
============================
cmd:CS
A stream - IO Link state is: DISCONNECTED
B stream -    Link state is: DISCONNECTED
C stream -    Link state is: DISCONNECTED
D stream -    Link state is: DISCONNECTED
E stream -    Link state is: DISCONNECTED
F stream -    Link state is: DISCONNECTED
G stream -    Link state is: DISCONNECTED
H stream -    Link state is: DISCONNECTED
I stream -    Link state is: DISCONNECTED
J stream -    Link state is: DISCONNECTED
K stream -    Link state is: DISCONNECTED
cmd:

CONNECTED TO ONLY ONE STATION
=============================
```

```
cmd: CS
A stream - IO  Link state is: CONNECTED to W6YEY
B stream -     Link state is: DISCONNECTED
C stream -     Link state is: DISCONNECTED
D stream -     Link state is: DISCONNECTED
E stream -     Link state is: DISCONNECTED
F stream -     Link state is: DISCONNECTED
G stream -     Link state is: DISCONNECTED
H stream -     Link state is: DISCONNECTED
I stream -     Link state is: DISCONNECTED
J stream -     Link state is: DISCONNECTED
K stream -     Link state is: DISCONNECTED
cmd:
```

If you are connected to several stations, the CSTATUS command shows your connect status as follows:

CONNECTED TO SEVERAL STATIONS
==========================

```
cmd: CS
A stream - I   Link state is: CONNECTED to W6YEY
B stream -     Link state is: CONNECTED to W6WNE
C stream -     Link state is: DISCONNECTED
D stream -     Link state is: CONNECT in progress
E stream -     Link state is: DISCONNECTED
F stream -     Link state is: CONNECTED to W6UOU via K7FT
G stream -     Link state is: DISCONNECTED
H stream - O   Link state is: DISCONNECTED
I stream -     Link state is: DISCONNECTED
J stream -     Link state is: DISCONNECTED
K stream -     Link state is: DISCONNECTED
cmd:
```

This sample display shows that:

CHANNEL O has the input and output channels - you are using it.

CHANNEL I is the last channel stream you heard a packet.

All other channels' states are shown as they might appear with multiple connections.

CTEXT
Command: CTEXT text

Mnemonic: CT
Default: Empty

Parameter: text Any combination of characters and spaces up to a maximum of 239 characters.

Refer to: CMSG, CMSGDISC, PASS

CTEXT is the 'automatic answer' text you type into a special location of the TNC's memory.

If CMSG is set to ON, the CTEXT message is sent as soon as another station connects to your station.

To type multiple-line CTEXT messages and include a carriage return (<CR>) character in your text, use the PASS character (<CTRL-V> is the default value) immediately preceding the carriage return (refer to the PASS command).

A typical CTEXT message might be:

I am not available right now <CTRL-V><CR>
Please leave your message, then disconnect <CR>

✅ **NOTE:** If you enter a text string longer than 239 characters, the following error message appears and the command is ignored:

```
?too long
cmd:
```

Use a parent sign (%), or an ampersand (&) as the first character in the CTEXT message to clear the previous message without having to type a **RESET** command.

The CTEXT message will not be sent if the **CMSG** command is set to OFF.

DISCONNE

Command: DISCONNE
Mnemonic: D
Default: Immediate Command

Parameter: none

Refer to: CONNECT, CONPERM

DISCONNE is an immediate command that initiates a disconnect request to the other station to which you are connected.

If you disconnect command is successful, your monitor will display:

*** *DISCONNECTED*^@

Other commands can be entered while a disconnect is in progress. New connections are not allowed until the disconnect is completed.

NOTES:

1. If the retry count is exceeded while you are waiting for the other station to acknowledge your disconnect command, your TNC automatically switches to the disconnected state.
2. If another disconnect command is entered while your TNC is trying to disconnect, the retry count is immediately set to the maximum number. In either case, your monitor displays:

*** *Retry count exceeded*
*** *DISCONNECTED*^@

Disconnect messaged are not displayed when your TNC is in Transparent Mode.

✅ **NOTE:** ^@ will not be displayed on the ordinary terminal.

DAYTIME date&time

Command: DAYTIME date&time
Mnemonic: DA
Default: clock not set

Parameter: date&time Current DATE and TIME to set.

Refer to: CONSTAMP, DAYUSA, MSTAMP, MBOD
DAYTIME sets the TNC's internal clock current date and time as well as the hardware Real Time Clock in the Pico. Once set, every second afterwards the TNC’s internal clock is updated by the values in the Pico’s Real Time Clock which is more accurate but only if the current year of the Real Time Clock does not match the year set in the TNC’s interncal clock. The best way to guarantee an update of the Real Time Clock is to set a date with a year that is 1 less than the current real date&time. Then set it again with the current real date&time .

The date&time parameter is used in the Packet Mode by the commands CONSTAMP and MSTAMP to 'time stamp' received and monitored messages.

Entries in the "heard" (displayed ny MHEARD are also time stamped if data&time has been set.

The clock is not set when the Pico TNC is first turned on or Reset. If so the **DAYTIME** command will display:

```
cmd: DAYTIME
Clock Not Set?
```

If the clock has been set and you type **DAYTIME** without a parameter, the TNC displays the current date and time information.

The format for entering the date&time is:

YYMMDDhhmmss

where:

YY is the last two digit of the year. (00-99).
MM is the two digit month code (01-12).
DD is date (01-31).
hh is the hour (00-23).
mm is the minutes after the hour (00-59)
ss is the second after the minute (00-59)

Entering the number 0-9 with leading zeros; all fields must be exactly two digits.The TNC will not echo the new string, thus you must issue DAYTIME command to confirm an entry.

Example:

```
cmd: DAYTIME 9210030935
cmd: DAYTIME
92/10/03 09:35:10
```

DAYUSA ON/OFF

Command: DAYUSA ON/OFF
Mnemonic: DAYU
Default: ON

Parameter: ON Date is displayed in a mm/dd/yy format.
OFF Date is displayed in a dd-mm-yy format.

Refer to: CONSTAMP, DAYTIME, MSTAMP

The DAYUSA command allows you to determines the format of the TNC's date display. If DAYUSA is set to ON, the standard United Sates format is used. If DAYUSA is set to OFF, the standard European format is used. The command affects the format of the date display that is used in "time stamps. It also affects the display when you enter a DAYTIME command without parameters.

If DAYUSA is set to ON, October 3, 1992 and 09:35:10 AM would be displayed as:

```
cmd:DAYUSA ON
DAYUSA was ON
cmd:DAYTIME
10/03/92 09:35:10
```

If DAYUSA is set to OFF, the same date and time would be displayed as:

```
cmd:DAYUSA OFF
DAYUSA was ON
cmd:DAYTIME
03-Oct-92 09:35:10
```

DELETE ON/OFF
Command: DELETE ON/OFF
Mnemonic: DEL
Default: ON

Parameter: ON The <DELETE> ($7F) key is used for editing your typing.
OFF The <BACKSPACE> ($08) key is used for editing your typing.

Refer to: BKONDEL, CANLINE

Use the **DELETE** command to select the key to use for deleting while editing.
Type the selected DEL key to delete the last character from the input line.
You cannot use the DEL key to delete text before the beginning of a line. Use the PASS character to delete <CR> characters that have been typed into the text.

The **BKONDEL** command controls how the TNC indicates deletion.

To see a corrected display of the current line after deleting characters, type the redisplay-line character, which is set by the **REDISPLA** command.

DIGIPEAT ON/OFF

Command: DIGIPEAT ON/OFF
Mnemonic: DIG
Default: ON

Parameter: ON The TNC will digipeat packets, if another station requests it.
OFF The TNC will not digipeat packets.

When this parameter is set to ON, any packet that is received with your TNC's call sign (including

SSID) in the digipeater list of its address field will be re-transmitted.

Each station that is included in the digipeater list relays the packet, in turn, and will make the packet so that it will not accidentally relay it twice (unless it is requested to do so.) In addition, all stations will relay the packet in the correct order. Digipeating takes place concurrently with other TNC operations and does not interfere with normal operation of a packet station.

Many stations set **DIGIPEAT** to ON most of the time. You may wish to set it to OFF if you are not at home, or if your transmitter/receiver relay makes enough noise to wake you up at night.

The **HID** command enables automatic re-transmission of identification packets when your station acts as a digipeater.

DISPLAY (class)
Command: DISPLAY (class)
Mnemonic: DISP
Default: Immediate Command

Parameter: class Optional parameter identifier, one of the following.

- (A)synchronous RS232C port parameters.
- (C)haracter display special characters.
- (H)ealth display health parameters.
- (I)d display ID parameters.
- (L)ink display Link parameters.
- (M)onitor display monitor parameters.
- (T)iming display timing parameters.

If you type DISPLAY without a class parameter, all control parameters and their current values are displayed. You can specify the optional parameter class to display subgroups of related parameters. To display an individual parameter, enter the parameter name without an option.

Example:

```
cmd: DISPLAY ASYNC  (or A)
```

All asynchronous port (RS-232C) parameters are displayed.

DWAIT n

Command: DWAIT n
Mnemonic: DW

Default:	16

Parameter:	n 0 - 250 specifies default wait time in 10 mS intervals.

**Refer to:	TXDELAY, AXDELAY, AXHANG**

DWAIT helps to avoid collisions with digipeated packets.

Unless the TNC is waiting to transmit digipeated packets, DWAIT forces the TNC to pause after last hearing data on the channel, for the duration of the DWAIT (Default Wait) time, before it begins its transmitter key-up sequence.

Wherever possible, the value of DWAIT should be agreed on by all stations in a local area when digipeaters are used in the area. The best value will be determined byexperimenting.

DWAIT is a function of the key-up time (TXDELAY) of the digipeater stations and helps alleviate the drastic reduction of throughput that occurs on a channel when digipeated packets suffer collisions.

DWAIT is necessary because digipeated packets are not retired by the digipeater, but are always restarted by the originating station. When all stations specify a default wait time, and right value of "n" is chosen, the digipeater captures the frequency every time it has data to send. - digipeated packets are sent without this delay.

TYPE OF OPERATION	TIME (mS)	DWAIT VALUE
Digipeaters	0	0
Local Keyboard	160	16
PBBS	320	32
File Transfer	480	48

ECHO ON/OFF

Command:	ECHO ON/OFF
Mnemonic:	E
Default:	ON

Parameter:	ON Characters received from the computer or terminal are echoed by the TNC.
 OFF Characters are not echoed.

The ECHO command controls local echoing by the TNC when it is in the Command or Converse Mode.   Local echoing is disable in the Transparent Mode.

Set ECHO ON if you do not see your typing appear on your display.

SET ECHO OFF if you see each character you typed doubled.

ECHO is set correctly when you see the characters you type display correctly.

ESCAPE ON/OFF

Command: ESCAPE ON/OFF
Mnemonic: ES
Default: OFF

Parameter: ON The <ESCAPE> character ($1B) is output as "$"($24).
OFF The <ESCAPE> character is output as <ESCAPE> ($1B).

Refer to: AFILTER, MFILTER

The **ESCAPE** command selects the character to be output when an <ESCAPE> character is to be sent to the terminal. The <ESCAPE> translation is disabled in Transparent Mode.

The **ESCAPE** character selection is provided because some computers and terminal emulators interpret the <ESCAPE>character as a special command prefix. Such terminals may alter their displays depending upon the characters following the <ESCAPE>.

Set **ESCAPE** ON if you have such a terminal to avoid unexpected text strings from other packeteers.

Refer to the **AFILTER** and **MFILTER** commands for information about character stripping (rather than character translation) in monitored packets.

FILE

Command: FILE
Mnemonic: F
Default: Immediate Command

Parameter: None

Refer to: "Using the Bulletin Board Feature" in the Operating Instruction Manual.

FILE command display the files or message in the BBS.

FLOW ON/OFF

Command: FLOW ON/OFF

Mnemonic: F
Default: ON

Parameter: ON Type-in flow control is active.
OFF Type-in flow control is not active.

Refer to: XFLOW, RSFLOW, FLOVER

When FLOW is ON, type-in flow control is active. Any character typed on your keyboard causes output from the TNC to the terminal to stop until any of the following conditions exist.

- A packet is forced (in the Converse Mode).
- A line is completed (in the Converse Mode).
- The packet length (See **PACLEN**) is exceeded.
- The terminal output buffer fills up.

Canceling the current command or packet or typing the redisplay-line character also causes output to resume. Type-in flow control is not used in the Transparent Mode.

Setting FLOW ON prevents inbound or received data from interfering with your keyboard data entry. If you (and the person you are talking to) normally wait for a packet from the other end before starting to respond, you can set FLOW OFF.

Some packet bulletin board programs (PBBS) may work best with FLOW set to OFF.

Some vintage computers such as the Radio Shack Color Computer with "software UARTs" may be unable to send and receive data at the same time. If you are using that type of computer, set FLOW to ON.

FRACK n

Command: FRACK n
Mnemonic: FR
Default: 3

Parameter: n 1 to 15, specifying frame acknowledgment time-out in one-second intervals.

FRACK is the FRame ACKnowledgement time in seconds that your TNC will wait for acknowledgement of the last-sent protocol frame before resending or "retiring" that frame.

After sending a packet requiring acknowledgement, the TNC waits for FRACK seconds time-out before incrementing the retry counter and sending the frame again. If the packet address includes digipeater instructions, the time between retries is adjusted to:

Retry interval = "n" x (2n x m +1)

where m is the number of intermediate relay stations.

When a packet is retried, a random wait time is added to any other wait times in use. This avoid lockups in which two packet stations repeatedly send packets which collide with each other.

Recommended value for the FRACK is as follows:

TYPE OF OPERATION	FRACK VALUE
Local Keyboard	3 (default)
Local Keyboard	6 (when the channel is congested)
PBBS	6
File Transfer	10

FULLDUP ON/OFF

Command: FULLDUP ON/OFF
Mnemonic: FU
Default: OFF

Parameter: ON Full duplex mode is ENABLED.
OFF Full duplex mode is DISABLED.

When full-duplex mode is disabled, the TNC makes use of the DCD (Data Carrier Detect) signal from its modem to avoid collisions; the TNC acknowledges multiple packets in a single transmission with a single acknowledgement.

When full-duplex mode is enabled, the TNC ignores the DCD signal and acknowledges packets individually.

Full-duplex operation is useful for full-duplex radio operation, such as through Sattelites AO-13 or JAS-1B. It should not be used unless both your station and the other station can operate in full-duplex.

You may also find full-duplex mode useful for some testing operations, such as analog or digital loopback tests.

HBAUD n

Command : HBAUD n
Mnemonic: HB
Default: 1200

Parameter: n Values specifying the rate or signaling speed in bauds from the TNC to the radio.

HBAUD sets the radio (on-air) baud rate only in the packet operating mode. HBAUD has no relationship to your computer terminal program,s baud rate.

✅ **NOTE:** On the Pico Tnc currently only 1200 baud is supported. There is future support planned for 300 baud to use for HF Packet. For now always set **HBAUD** to 1200. The TNC will accept other numbers, but the TNC's modem port is not acceptable of using anything other than 1200 baud.

Example:HBAUD 1200

HEADERLN ON/OFF

Command: HEADERLN ON/OFF
Mnemonic: HE
Default: OFF

Parameter: ON The header for a monitored packet is printed on a separate line from the packet text.
OFF The header add the text of monitored packets are printed on the same line.

Refer to: MRPT, MSTAMP

HEADERLN affects the display of monitored packets. When **HEADERLN** is OFF, the address information is shown on the same line as the packet text:

W6YEY>W6UOU: Go ahead and transfer the file.

When HEADERLN is ON, the address is shown, followed by a <CR><LF>that puts the packet text on a separate line:

W6YEY>W6UOU:
Go ahead and transfer the file

If **MRPT** or **MSTAMP** are ON, set **HEADERLN** ON; long headers may extend across your screen or page when these functions are active.

HEALLED ON/OFF

Command: HEALLED ON/OFF
Mnemonic: HEAL
Default: OFF

Parameter: ON Automatically runs diagnostic tests on the microprocessor when you turn the

unit on. If all checks are satisfactory, it alternately lights the CON and STA LEDs.
OFF Does not run diagnostic test and no LEDs will light.

Allows you to make a quick check of the Virtual TNC's operation. If the LEDs do not flash, the Pico's Virtual TNC's internal software has failed. This needs to be tested.

HID ON/OFF

Command: HID ON/OFF
Mnemonic: HI
Default: OFF

Parameter: ON Your TNC sends HDLC identification as digipeater.
OFF Your TNC does not send HDLC identification.

Refer to: UNPROTO, MYCALL, ID, MYALIAS

The HID command activates or disables your TNC's automatic periodic transmission of identification packets when it is operating as digipeater. This identification consists of an un-sequenced I-frame with your station identification (MYCALL) and MYALIAS in the data field.

Set HID ON to force your TNC to send an ID packet every 9.5minutes when it is being used as digipeater.

The HID identification packet is addressed to the "CQ" address set by the UNPROTO command.

Your station identification is the call sign you have set with the MYCALL command, with "digipeater" appended.

W6YEY >ID: W6WNE/R

✅ **NOTE:** You cannot change the 9.5-minute automatic interval timing.

ID

Command: ID
Mnemonic: I
Default: Immediate Command

Parameter: none

Refer to: UNPROTO, MYCALL, HID

ID is an immediate command that sends a special identification packet. The ID command allows you to send a final identification packet when you take your station off the air.

Note that **HID** must set on. ID forces a final identification packet to be sent when a digipeater station is being taken off the air. The identification consists of an unnumbered I-Frame, with its data field containing **MYALIAS** (if any) and your **MYCALL** station identification and characters "/R".

The ID identification packet is sent only if the digipeater has transmitted since the last automatic identification.

K

Command: K
Mnemonic: K
Default: Immediate Command

Parameter : none

Refer to: Same as CONVERSE. Refer to CONVERSE

KILL

Command: KILL
Mnemonic: KI
Default: Immediate Command

Refer to: "Using the Bulletin Board Feature" in the Operating Instruction Manual.

KILL command erases the file or message the BBS.

LCOK ON/OFF

Command: LCOK ON/OFF
Mnemonic: L
Default: ON

Parameter: ON Causes the TNC to send lower case characters to the computer or terminal.
OFF Causes the TNC to translate lower case characters to upper case.

If LCOK is OFF, lower case characters are translated to upper case before they are sent to the terminal. Input characters and echo are not translated.

✅ **NOTE:** Case translation is disabled in Transparent Mode.

If your computer or erminal does not accept lower case characters, such as the APPLE II, it may react strangely when the TNC sends them to it. The LCOK command allows you to translate all lower case characters that are received in packets, as well as messages from the TNC, to upper case.

HINT: Since echoes of the characters you type are not translated to uppercase, you can use this command to make your display easier to read when you converse in a connected mode. If you and the other station set LCOK to OFF, you can each type your own message in lower-case, and the other station's incoming packets will be in upper case. This makes it easier to distinguish between incoming and outgoing packets.

LCALLS call1, call2..call8

Command: LCALLS call1, call2..call8
Mnemonic: LCA
Default: Empty

Parameter: call Call sign list. Up to 8 calls, separated by commas.

Refer to: BUDLIST

LCALLS uses arguments to determine how your TNC monitors the packet channels and displays information - which station's packet will be displayed.

Each call sign may include an optional SSID. **LCALLS** works with **BUDLIST** and it allows selective monitoring of other packet stations. **LCALLS** and **BUDLIST** determine which packets will be displayed when you have set **MONITOR** ON.

BUDLIST specifies whether the call signs in the list are the stations that you want to ignore or, alternatively, they are the only stations that you want to monitor. If you want to monitor only for packets from a limited list, you may enter your selected stations in the **LCALLS**, and set **BUDLIST** to ON.

If you want to ignore packets from a limited list, you may list the call signs to ignore in the **LCALLS** and set **BUDLIST** to OFF.

LCSTREAM ON/OFF

COMMAND: LCSTREAM ON/OFF
Mnemonic: LCS
Default: ON

Parameter: ON Cause the character that immediately follows the STREAMSWITCH character to be changed uppercase, before it is acted upon.
OFF Causes the character that immediately follows the STREAMSWITCH character to be acted upon normally.

Refer to: STREAMSW

When **LCSTREAM** is set to ON the character that immediately follows the STREAMSWITCH character is changed to upper case before it is acted upon as a parameter to that command. Since you can specify the stream you desire with either an uppercase or a lower-case letter, the **LCSTREAM** command simplifies management of multiple connections.

When **LCSTREAM** is set to OFF, the case of the character that immediately follows the STREAMSWITCH character is important.

✅ **NOTE:** A lower-case letter will result in an error message.

LFADD ON/OFF

Command: LFADD ON/OFF
Mnemonic: LF
Default: OFF

Parameter: ON A <LF> character is added to outgoing packets following each <CR> transmitted in the packet.
OFF No <LF> is added to outgoing packets.

Refer to: AUTOLF

LFADD is similar to **AUTOLF**, except the line feed characters are added to outgoing packets instead of the displayed text. This feature helps make your TNC compatible with other packet radio TNCs.

Set **LFADD** to ON if the station you are communicating with reports that your packets are being overprinted.

✅ **NOTE:** This character is disabled in the Transparent Mode.

LFIGNORE ON/OFF

Command: LFIGNORE ON/OFF
Mnemonic: LFI
Default: OFF

Paramete: ON Causes the TNC to ignore <LF> characters it receives from another station.

OFF Causes the TNC to print any <LF> characters it receives.

Set LFIGNORE to ON if you feel you are receiving too many line feeds from another station.

NOTE: This command has no effect in the Transparent Mode.

MONITOR ON/OFF

Command: MONITOR ON/OFF
Mnemonic: M
Default: ON

Parameter: ON Packet activity is monitored.
OFF Packet activity is not monitored.

Refer to: BUDLIST, LCALLS, MALL, MCON, MRPT, MSTAMP, MHEADERLN

If **MONITOR** is set to ON and the TNC is not in the Transparent Mode, packets not addressed to you are displayed. The addresses in the packet are displayed along with the data portion of the packet. For example"

W6YEY > W6UOU-7: I am ready to transfer the file.

The call sign are separated by a ">" and the substation ID field (SSID) is displayed if it is other than 0. The **MALL**, **BUDLIST**, and **LCALLS** commands determine which packets are to be monitored. The **MCON** command controls the action of the monitor mode when the TNC is in a connected state.

All monitor functions are disabled in the Transparent Mode.

HEADERLEN controls the format of the monitor display. If you wish to see the station address on a separated line from the text, set **HEADERLN** ON. **MRPT** enables monitoring of the digipeater routing as well as source and destination address for each packet.

MSTAMP includes a time stamp with the address if you have set **DAYTIME**.

MALL ON/OFF

Command: MALL ON/OFF
Mnemonic: MA
Default: ON

Parameter: ON Both connected and non-connected packets are monitored.
OFF Only non-connected packets are monitored.

Refer to: **MONITOR, BUDLIST, LCALLS, MTCALL, MRCALL, MTR**

The **MALL** command determines the class of monitored packets. When MALL is set to OFF, only packets from unconnected stations determined by **BUDLIST**, **LCALLS**, are displayed. This is the normal setting when you are communicating to group of unconnected stations.

When MALL is set to ON, all eligible frames are displayed, including those that are sent between two other connected stations.

The MALL command is handy for diagnostic purpose or for general monitoring.

MAXFRAME n

Command: MAXFRAME n
Mnemonic: MAX
Default: 4

Parameter: n 1 to 7 signifying a number of packet frames.

Refer to: **PACLEN, FULLDUP**

MAXFRAME sets an upper limit on unacknowledged packets your TNC permits on the radio link at any one time. **MAXFRAME** also sets the maximum number of contiguous packets your TNC will send during any given transmission.
If some but not all of the outstanding packets are acknowledged, a smaller number may be transmitted the next time, or new frames may be included in the re-transmission so that the total number of unacknowledged packet frame does not exceed "n".

The best value of MAXFRAME depends upon your local channel conditions. In most case of keyboard-to-keyboard direct or local operation (link that do not require going through digipeaters), you can use the default value MAXFRAME 4.

When the amount of packet traffic, the path in use, the digipeaters involved, or other variables not under your control make packet operation difficult (as shown by lots of retries!), you can actually improve your throughput by reducing MAXFRAME.

If packet traffic is heavy or the path is poor, reduce MAXFRAME to 3 or 2.

If you are sharing the channel with several PBBSs and digipeaters, or when you are working a PBBSs or other type of host communications, reduce MAXFRAME to 1.

If the radio link is good, an optimal relationship exists between the parameters set by these commands, so that the maximum number of characters outstanding does not exceed the receive buffer space of the TNC receiving data.

MBOD ON/OFF

Command: MBOD ON/OFF
Mnemonic: MB
Default: OFF

Parameter: ON Enable the message (bulletin) board.
OFF Disable the message (bulletin) board.

Refer to: MYMCALL, DAYTIME

This TNC has built-in bulletin board capability. To use this feature, you must set **MBOD** to ON. Before you do this, however, make sure that you have entered an exclusive call sign in **MYMCALL**.

✅ **NOTE:** To use the bulletin board feature, you must enter a call sign into **MYMCALL**.

MCON ON/OFF

Command: MCON ON/OFF
Mnemonic: MC
Default: OFF

Parameter: ON Monitors packets from all other stations while your TNC is connected to another station.
OFF Only monitors packets from the station you are connected to.

Refer to: MALL, MONITOR

Turn **MCON** ON to monitor other stations on the frequency while your TNC is connected to another station.

MCOM ON/OFF

Command: MCOM ON/OFF
Mnemonic: MCOM
Default: OFF

Parameter: ON Connected, disconnected, information, UA (Unnumbered Acknowledgement), and DM (busy signal) frames are monitored.
OFF Only information frames are monitored.

Refer to: **MALL, MONITOR**

When **MCON** is set to ON, both connected and disconnected frames are displayed, if **MONITOR** is also set to ON. Connected, disconnected, information, UA and DM frames are indicated by <C>, <D>, <I>, <UA>, and <DM>, respectively. Like other monitor commands, stations that are monitored are determined by **BUDLIST** and **LCALLS**.

Set **MCOM** to OFF to display only I frames (packets that contain user information).

MFILTER n1[, n2[,n3[,n4]]]

Command: MFILTER n1[, n2[,n3[,n4]]]
Mnemonic: MFI
Default: $00

Parameter: 0 to $7F (0 to 127 decimal) specifies an ASCII code. Up to four characters may be specified.

Refer to: **AFILTER**

Use **MFILTER** to select characters to be "filtered", or excluded from monitored packets. Parameters "n1," "n2", etc., are the ASCII codes for the characters you want to filter. You can enter up to four characters in either hex or decimal.

1. To prevent a <CTRL-L> character from learing your screen, set MFILTER 12.
2. To eliminate <CTRL-Z> character, which some computer interpret as end-of-file markers, set MFILTER 26.
3. To eliminate <CTRL-G> characters, which beep your computer or terminal, set MFILTER 7.

MHEARD

Command: MHEARD
Mnemonic: MH
Default: Immediate Command

Parameter: none

Refer to: **MSTAMP, MHCLEAR**

MHEARD is an immediate command that displays a list of stations heard since the last time the MHEARD buffer was cleared.

Use the **MHC** command to clear the MHEARD buffer.

The maximum number of heard stations that can be logged is 18. If more stations are heard, earlier entries are discarded.

NOTES:

1. Stations that are heard directly are not marked with a * in the heard log.
2. Stations that have been repeated by a digipeater are marked.
3. If you clear the list of stations heard at the beginning of a session, you can use this command to keep track of the stations that are active during that period.
4. Logging of stations heard is disabled when **PASSALL** is ON.

When **DAYTIME** has been used to set the date and time, entries in the heard log are date and time stamped.

MHCLEAR

Command: MHCLEAR
Mnemonic: MHC
Default : Immediate Command

parameter: none

Refer to: MHEARD

The **MHCLEAR** command clears the list of stations heard. You can use this command together with the **MHEARD** to keep track of any stations heard over a particular period of time, such as evening or a week. Be sure to clear the list of stations heard when you first begin to monitor packet activity.

MRPT ON/OFF

Command: MRPT ON/OFF
Mnemonic: MR
Default: ON

Parameter: ON Show digipeaters in the header; stations heard directly are marked with asterisk.
OFF Show packets only from originating and destination stations.

Refer to: HEADERLN

MRPT affects the way monitored packets are displayed. When **MRPT** is OFF, only packets from the originating station and the destination station are displayed:

```
W6YEY*> W6UOU-7<I;0,3>:
```

In addition, while **MRPT** is OFF, the TNC will remove all indications of digipeater paths in connect requests, and all connect and disconnect stamps.

When MRPT is ON, the call signs of all stations in the entire digipeat path are displayed. The call sign of the stations received directly are flagged with an asterisk (*):

```
W6YEY*>W6UOU-7>W6WNE>K7FT <I;0,3>:
```

MSTAMP ON/OFF

Command: MSTAMP ON/OFF
Mnemonic: MS
Default: OFF

Parameter: ON Monitored frames ARE time stamped
OFF Monitored frames ARE NOT time stamped.

Refer to: HEADERLN, DAYTIME, DAYUSA, CONSTAMP

The **MSTAMP** command activates or disables time stamping of monitored packets. When your TNC's internal clock is set, date and time information is available for automatic logging of packet activity or other applications.

When **MSTAMP** is OFF, the packet header display looks like this:

```
W6YEY-7>W6UOU-7>W6WNE <I,2,2>
```

When **MSTAMP** is ON, the display looks like this:

```
09/10/25 22:10:45 W6YEY-7>W6UOY-7>W6WNE <I,2,2>
```

Set the date and time with the **DAYTIME** command.

Setting **MSTAMP** ON increase the length of the address display.

Set **HEADERLN** ON to display this information on a separate line.

MYCALL call (-n)

Command: MYCALL call (-n)
Mnemonic: MY
Default: NOCALL

Parameters: call your call sign n = 0 to 15, indicating an optional substation ID (SSID)

Refer to: MYALIAS, ID, HID, MYMCALL, CONNECT, XMITOK

Use the MYCALL command to load your call sing into your TNC's RAM. Your call sign is inserted automatically in the FROM address field for all packets originated by your TNC. MYCALL is also used for identification packets (see HID and ID).

Your TNC accepts connect request frames with your MYCALL in the TO field and repeats frames with this call sign in the digipeated field.

NOTE: The factory default is set to "NOCALL" and you cannot send any data until you set to your own unique ID.

Two or more stations cannot use the same call sign (including SSID) on the air at the same time.

Use SSID to distinguish between two stations with the same call sign.

The SSID will be zero(0) unless explicitly set to another value.

Although there is no standardization of SSIDs at present, most packet operators use SSID 0 (zero) for manual or local keyboard operation of their main station, and an SSID of (-1) for their **MYMCALL** bbs call and or (-2) for a secondary station or dedicated digipeater under their responsibility.

Local area networks operated or maintained by a packet group or club may use the same call sign for several stations in their network, each node or unit being identified with a deferent SSID.

MYALIAS call (-n)

Command: MYALIAS call (-n)
Mnemonic: MYA
Default: Empty

Parameter: call Alternate identity of your TNC. n = 0 to 15, an optional substation ID (SSID).

Refer to: MYCALL, ID, HID, DIGIPEAT, MYMCALL

MYALIAS specifies an alternate call sign (in addition to the call sign specified in **MYCALL**) for use as a digipeater only.

✅ **NOTE:** The TNC will not allow other stations to connect to your MYALIAS call sign.

MYALIAS permits both normal HID identification and an alias alternate, repeater-only "call sign."

Wide-coverage digipeater operators in some areas change their call sign to a shorter and (usually) easier to remember identifier.

Identifiers used include International Civil Aviation Organization (ICAO) airport IDs, sometimes combined with telephone area codes.

NEWMODE ON/OFF

Command: NEWMODE ON/OFF
Mnemonic: NE
Default: OFF

Parameter: ON At connect time, the TNC automatically goes to the mode specified by CONMODE and return to the Command Mode at disconnect.
OFF At connect time, the TNC automatically goes to the mode specified by CONMODE as soon as the link is established. The TNC does not return to the Command Mode at disconnect.

Refer to: CONMODE, NOMODE, CONNECT

NEWMODE determines how the TNC behaves when the link is broken. The TNC always switches to a mode specified by the **CONMODE** at the time of connection, unless **NOMODE** is ON.

Set **NEWMODE** for the type of operation most suitable to your needs.

If **NEWMODE** is OFF and the link is disconnected, the TNC remains in converse or transparent mode unless you have forced it to return.

When **NEWMODE** is ON and the link is disconnected, or if the connect attempt fails, the TNC returns to the Command Mode.

NOMODE ON/OFF

Command: NOMODE ON/OFF
Mnemonic: NO
Default: OFF

Parameter: ON The TNC does not switch the mode specified by CONMODE and NEWMODE automatically.

OFF The TNC switches to the mode specified by CONMODE and NEWMODE automatically.

Refer to: CONMODE, NEWMODE

When **NOMODE** is ON, the TNC never switches from Converse or Transparent mode to command mode (or vice verse) by itself. Only specific commands (CONVERSE, K, TRANS, or <CTRL-C>) typed by you can change the operating mode.

When **NOMODE** is OFF, the TNC switches modes automatically according to the way NEWMODE and CONMODE are set.

NUCR ON/OFF

Command: NUCR ON/OFF
Mnemonic: NU
Default: OFF

Parameter: ON <NULL> characters ARE sent to the terminal following <CR> characters.
OFF <NULL> characters are not sent to the terminal following <CR> characters.

Refer to: NULF, NULLS

Some of the older electromechanical terminals (Teletype machines) and printer terminals require some extra time for the printing head to do a line feed and return to the left margin. NUCR ON solves this problem by making the TNC send <NULL> characters (ASCII code $00) to your computer or terminal. This introduces any necessary delay after any <CR> sent to the terminal.

The NULLS command sets the number of individual <NULL> characters that are to be sent when NUCR is ON.

NOTE: Set NUCR ON if your terminal or printer misses one or more characters after responding to a <CR>. If this is the case, you will sometimes see overtyped lines.

NULF ON/OFF

Command: NULF ON/OFF
Mnemonic: NUL
Default: OFF

Parameter: ON <NULL> characters are sent to the terminal following <LF> characters.

OFF <NULL> characters are not sent to the terminal following <LF> characters.

Refer to: **NUCR, NULLS**

NULLS n

Command: NULLS n
Mnemonic: NULL
Default: 0

Parameter: n 0 to 30 specifies the number of <NULL> characters to be sent to your computer or terminal after <CR> or <LF> when NUCR or NULF are set ON

Refer to: **NUCR, NULF**

NULLS specifies the number of <NULL> characters (ASCII code $00) to be sent to the terminal after <CR> or <LF> is sent.

NUCR and/or NULF must be set to indicate whether nulls are to be sent after <CR>, <LF>, or both.

Devices requiring null after <CR> are typically hard-copy devices requiring time for carriage movement. Devices that require null after <LF> are typically CRTs that scroll slowly.

The null characters are sent from your TNC to your computer only in the Converse and Command Modes.

PACLEN n

Command: PACLEN n
Mnemonic: PACL
Default: 128

Parameter: n 0 to 255 specifies the maximum length of the data portion of a packet.
0 Zero is equivalent to 256

Refer to: **MAXFRAME**

PACLEN sets the maximum number of user data bytes to be carried in each packet's information field. "User data" means the characters you actually type at your keyboard (or send form a stored file).

Your TNC automatically transmits a packet when the number of characters you type (or send from disk) for a packet equals "n".This value is used in both the Converse and Transparent Modes.

Most keyboard-to-keyboard operators use the default value of 128 bytes for routine VHF/UHF packet service.

Experiment with deferent values for **MAXFRAME** and **PACLEN** to find the combination best suited to your operating conditions, especially if you are transferring files.

The lower the value of **PACLEN**, the greater the possibility of getting packets though the link without "hits" or retries.

Increase **PACLEN** to 256 bytes if transferring files to a nearby station over a high quality path.

Reduce **PACLEN** to 64, or even 32 when working "difficult" or marginal radio path.

If the radio link is good, an optimal relationship will exist between the parameters set by these commands. Set **PACLEN** so that the maximum number of characters outstanding does not exceed the receive buffer space of the TNC receiving data.

✅ **NOTE:** It is not necessary that two TNCs be set to the same PACLEN value to exchange data; some TNCs, however, may not be compatible when frames contain more than 128 data characters.

PACTIME EVERY/AFTER n

Command: PACTIME EVERY/AFTER n
Mnemonic: PACT
Default: AFTER 10

Parameter: n 0 to 250 specifies 100 millisecond intervals.
EVERY Packet time-out occurs every "n" times 100 mS.
AFTER Packet time-out occurs when "n" times 100 mS. elapse without input from the computer or terminal.

Refer to: CPACTIME, TRANS

A **PACTIME** parameter is always used in the Transparent Mode.**PACTIME** is also set in the Converse Mode if **CPACTIME** is on.

When EVERY is specified, the characters you type or send from disk are packetized and queued for transmission every "n" times 100 mS.

When AFTER is specified, the characters you type or send from disk are packetized when input from the terminal stops for "n" times 100 mS.

A zero-length packet will never be produced. The timer is not started until the first character or byte is entered.

A value of 0(zero) for "n" is allowed; zero means packets are sent with no wait time.

PARITY n

Command: PARITY n
Mnemonic: PAR
Default: 3

Parameter: n 0 to 3 selects a parity option from the table below.

Refer to: AWLEN, 8BITCONV, RESTART

PARITY sets the Pico TNC's data parity for terminal or computer data when using the TTL serial port according to the following table:

0 = no parity
1 = odd parity
2 = no parity
3 = even parity

The parity bit, if present, is stripped automatically on input, and is not checked in the Command and Converse Modes.

In the Transparent Mode all eight bits (including parity) are transmitted in packets. When "no parity" is set and AWLEN is 7, the eighth bit is set to 0 (zero).

✅ **NOTE:** Parity does not affect the USB serial port on the Pico TNC

PASS n

Command: PASS n
Mnemonic: PAS
Default: $16 <CTRL-V>

Parameter: n 0 to $7F(0 to 127 decimal) specifies an ASCII character code.

Refer to: BTEXT, CTEXT, DTEXT, MTEXT

PASS selects the ASCII character used for the "pass" input editing command.

The Parameter "n" is the numeric ASCII code for the character you will use to signal that the character immediately following it is to be included in a packet or text string.

✅ **NOTE:** You can enter the code in either hex or decimal.

Use the **PASS** character (default is <CTRL-V>) to send characters that usually have special functions.

A common use for the pass character is to allow <CR> to be included in the BTEXT, CTEXT, DTEXT and MTEXT messages so that the transmitted information appears on several short lines rather than a single longer line. Use the PASS character to insert <CR> at the end of a short line:

Example:

```
BT Notice: <CTRL-V><CTRL-M>
Meeting at the Firehouse tonight <CTRL-V><CTRL-M>
at 8:00 PM. All welcome! <CR>
```

The other station's monitor shows:

```
Notice:
Meeting at the Firehouse tonight
at 8:00 PM. All welcome !
```

Without the PASS character, the,message would probably look this:

```
Notice: Meeting at the Firehouse tonight at 8:00 PM.  All
welcome!
```

In like manner, you can include <CR> in text when you are in the Converse Mode to send multiline packets. (The default send-packet character is <CR>.)

PASSALL ON/OFF

Command: PASSALL ON/OFF
Mnemonic: PASSA
Default: OFF

Parameter: ON Your TNC will accept packets with invalid CRCs (Cyclic Redundancy Check).
OFF Your TNC will only accept packets with valid CRCs.

Refer to: MHEARD, RETRY

PASSALL permits the TNC to display packets received with invalid CRC fields; the error-detecting mechanism is turned off.

Packets are accepted for display despite CRC errors if they consists of an even multiple of eight bits and are up to 330 bytes. The TNC attempts to decode the address field and displays the call sign(s) in the standard monitor format, followed by the text of the packet.

PASSALL is normally turned off, therefore, the protocol ensures that received packet data is error-free by rejecting packets with invalid CRC fields.

PASS (sometimes called "Garbage Mode") may be useful for testing a marginal RF link or during operation under other unusual conditions or circumstances.

When you set PASSALL ON, while monitoring a moderately noisy channel "packets" are displayed periodically because there is no basis for distinguishing between actual packets and random noise.

✅ **NOTE:** When PASALL is ON, logging of station heard (for display by **MHEARD**) is disabled; the call sign detected may be incorrect.

PERSIST n

COMMAND: PERSIST n
Mnemonic: PE
Default: 127

Parameter: n 0 to 255 specifies the threshold value for random number attempt to transmit.
0 Signifies a 1/256th chance of transmitting every SLOTTIME.
255 Causes the TNC to transmit right away without delay.

Refer to: PPERSIST, SLOTTIME

The **PERSIST** command works with the **SLOTTIME** command to achieve true p-persistent CSMA (Carrier-Sense Multiple Access) in the KISS TNC mode and in AX.25 operation.

However, no real advantage will be obtained in AX.25 operation unless the order stations on the channel are also using PERSIST and **SLOTTIME**.

When the host (your computer) has queued data for transmission, the TNC monitors DCD (Data Carrier Detect) signal from its internal modem. The TNC waits indefinitely for DCD to go inactive.

When the Channel is clear, the TNC generates a random number between 0 and 255. If this number is less than or equal to "P" the TNC keys the radio's PTT line, wait 0.01 x TXDELAY seconds, and then transmits all frames in the queue. The TNC then unkeys the PTT line ad returns to the idle state.

If the random number is greater than "P", the TNC wait 0.01 x SLOTTIME seconds and repeats the procedure. If the DCD signal has gone active during this wait time, the TNC waits for DCD to clear before it continues.

The TNC waits an exponentially-distributed random interval after it senses that the channel is clear before it tries to transmit. When PERSIST and SLOTTIME are carefully set, several stations sending traffic are much less likely to collide with each other when they simultaneously detect that the channel is clear.

✅ **NOTE:** P=255 directs the TNC to transmit as soon as possible, regardless of the random number.

PPERSIST ON/ OFF

Command: PPERSIST ON/ OFF
Mnemonic: PP
Default: OFF

Parameter: ON The TNC uses PERSIST and SLOTTIME parameters when it executes p-persistent CSMA (Carrier Sense Multiple Access).
OFF The TNC uses DWAIT for TAPR-type 1 persistent CSMA.

Refer to: PERSIST, SLOTTIME

When **PPERSIST** is set to ON, the TNC uses the **PERSIST** and **SLOTTIME** parameters for p-persistent CSMA instead of normal TAPR-type DWAIT procedure to achieve CSMA operation.
Set PPERSIST to ON if you operate the frequency has many PBBSs activity.

READ

Command: READ
Mnemonic: R
Default: Immediate Command

Parameter: none

Refer to: "Using the Bulletin Board Feature" in the Operating Instruction Manual.

READ command reads BBS messages.

RETRY n

Command: RETRY n
Mnemonic: RE
Default: 10

Parameter: n 0 to 15 specifies the maximum number of packet retries.

Refer to: FRACK, CONPERM, TRIES

The AX.25 protocol uses retries - re-transmission of frames that have not been acknowledged. Frames are re-transmitted "n" times before the link is disconnected. The time between retries specified by the command FRACK. A value of 0 for "n" specifies an infinite number of retries.

RECONNECT call1[VIA call2[,call3...call9]]

Command: RECONNECT call1[VIA call2[,call3...call9]]
Mnemonic: REC
Default: Immediate command

Parameter: call1 This is the call sign of the station you are trying to reconnect to.
call2 Optional call sign(s) or station(s) you are attempting to digipeat through. You can use up to eight digipeat stations.

Refer to: CONNECT, DISCONNE

The **RECONNECT** command allows you to change the path you are using to communicate with another station.

✅ **NOTE:** This command works only when you are already connected to the station you are attempting to reconnect to.

When you use this command, any frames that may be in process between your station and the station you are reconnecting to may be lost.

REDISPLA n

Command: REDISPLA n
Mnemonic: RED
Default: $12 <CTRL-R>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII character code.

Refer to: FLOW

REDISPLA changes the redisplay-line input editing charactercode.

Parameter "n" is the numeric ASCII code for the character you will use when you want to re-display the current inputline.

NOTE: You can enter the code in either hex or decimal.
Type the **REDISPLA** character to re-display a line you have just typed. The following things will happen:

1. Type-in flow control is temporarily turned off (if it had been active). Any incoming packets that are pending are displayed.

2. A <BACKSLASH> is appended to the line you just typed and the line is shown below it. Only the final form of the line is shown if you have deleted or changed any characters.

3. You can now continue typing where you left off.

Use the redisplay-line character to see a "clean" copy of your input if you are using a printing terminal and you have deleted characters.

If **BKONDEL** is set OFF, deletions are designated with <BACKSLASH> characters, rather than by trying to correct the input line display. The re-display line is the corrected text.

Use the **REDISPLA** character if a packet is received while you are typing a message in the Converse Mode. You can see the incoming message before you send your packet without canceling your input.

RESPTIME n

Command: RESPTIME n
Mnemonic: RES
Default: 5

Parameter: n 0 to 250 specifies 100 millisecond intervals.

Refer to: DWAIT

RESPTIME adds a minimum delay before the TNC sends acknowledgment packets. This delay may run concurrently with the default wait time set by **DWAIT** and any random wait in effect.

Use **RESPTIME** delay to increase throughput during operations such as file transfer. When the sending the TNC usually sends the maximum number of full-length packets.

Occasionally, the sending TNC many not have a packet ready in time to prevent transmission from being stopped temporarily, with the result that the acknowledgement of earlier packets collides with the final packet of the series.

These collisions can be avoided if the receiving TNC sets RESPTIME to 10.

RESET

Command: RESET
Mnemonic: RESET
Default: Immediate command

Parameter: none

Refer to: RESTART, RAMTEST

RESET is an immediate command that resets all parameters to the TNC's PROM default settings and reinitializes the TNC.

WARNING!: All personalized parameters are monitor lists are lost.

To reinitialize the TNC using the current parameter values, turn the TNC off then on, or use the **RESTART** command instead.

WARNING!: Do not confuse RESET with RESTART

RESTART

Command : RESTART
Mnemonic: RESTART
Default: Immediate Command

Parameter : none

Refer to: RESET, RAMTEST

RESTART is an immediate command that reinitializes the TNC using the settings stored in the TNC RAM. Your personalized parameter settings are retained unchanged, except data stored in MHEARD.

The effect of the **RESTART** command is the same as turning the TNC OFF, then ON again.

RESTART does not default the parameter values in RAM. See the **RESET** command.

RXBLOCK ON/OFF

Command: RXBLOCK ON/OFF
Mnemonic: RX
Default: OFF

Parameter: ON Direct the TNC to place a $FF mark in front of received packets before they are printed on your terminal.
OFF Does not place a $FF mark in front of received packets.

The $FF mark makes easier to identify incoming packets.

When RXBLOCK is ON, your TNC automatically places a $FF mark in front of all incoming packets. The order of identity is:

```
 ------------------------------------------------------
|  $FF   |   LH    |   LL   |   PID   |  DATA UNIT    |
 ------------------------------------------------------
(PREFIX )  (LENGTH)        (PID)           (DATA)
```

where :

$FF is the header identify block (1 byte).

LH is the block length high byte (1 byte, high 4 bit, 111 fixed).

LL is the block length low byte (1 byte).

PID is the protocol ident of the control field of the received packet.

DATA UNIT is the contents of the received packet.

✅ **NOTE:** When use this command, check **AUTOLF** and **MFILTER** settings.

SCREENLN n

Command: SCREENLN n
Mnemonic: S
Default: 0

Parameter: n 0 to 255 specifies the screen or platen width in characters, of the terminal.

The SCREENLN command allows you to properly format your terminal's output. A <CR>, <LF> sequence is sent to the terminal at the end of a line in the Command and Converse modes after "n" characters have been printed. An "n" value of zero inhibits this function.

✅ **NOTE:** If your computer automatically formats output lines, set SCREENLN 0 to avoid a conflict between the two line formats.

SENDPAC n

Command: SENDPAC n
Mnemonic: SE
Default: $0D<CTRL-M>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII character code.

Refer to: PACLEN, CR, CPACTIME

Use the **SENDPAC** command to select the character used to force a packet to be sent in the Converse Mode. The parameter "n" is the numeric ASCII code for the character you want to use to force your input to be packetized and queued for transmission. You can enter the code in either hexadecimal or decimal numbers.

Use default **SENDPAC** value $0D for ordinary conversation with **CR** ON to send packets at natural intervals with <<CR>s> included in the packet.

When you are setting **CPACTIME** ON, set **SENDPAC** to some value not ordinarily used (say,

<CTRL-A>, with **CR** OFF). This setting forces packets to be sent without extra <CR> characters being sent in the text.

SLOTTIME n

Command: SLOTTIME n
Mnemonic: SL
Default : 10

Parameter: n 0 to 250 specifies the time interval during which the TNC waits between generating random numbers to see if it can transmit.

Refer to: PERSIST, PPERSIST

The **SLOTTIME** command works with the **PPERSIST** and **PERSIST** command to achieve true p-persistent CSMA (Carrier Sense Multiple Access) in KISS TNC mode and normal AX.25 operation. No real advantage, however, is obtained during AX.25 operation unless other stations on the channel are also using **PERSIST** and **SLOTTIME**.

START

Command: START n
Mnemonic: STA
Default: $11 <CTRL-Q>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII characters code.

Refer to: STOP, XFLOW, TRFLOW, XOFF, XON, RSFLOW

Use the START command to choose the User Start character you want to use to restart output from the TNC to the terminal after you have halted it by typing the User Stop character.

NOTES:

1. The User Stop character is set by the STOP command.

2. You can enter the value in either hex or decimal

If the User Start and User Stop characters are set to $00, software flow control to the TNC is disabled; the TNC will only respond to hardware flow control (CTS/RTS).

✅ **NOTE:** Hardware flow control is not available on the Pico TNC TTL or USB Serial Ports.

If the same character is used for both the User Start and User Stop characters, the TNC alternately starts and stops transmission on receipt of the character ("toggles").

STOP n

Command: STOP n
Mnemonic: STO
Default: $13 <CTRL-S>

Parameter: n 0 to $7F (0 to 127 decimal) specifies an ASCII character code.

Refer to: START, XFLOW, TRFLOW, XOFF, XON, RSFLOW

Use the STOP command to select the User Stop character you want to use to stop output from the TNC to the terminal. Type this character to halt the TNC's output to your monitor so that you can read the received text before it scrolls off your screen display.

NOTES:

1. Output is restarted with the User Start character.
2. The User Start character is set by the START command.
3. You can enter the value in either hex or decimal.

STREAMSW n

Command: STREAMSW n
Mnemonic: STR
Default: $01 <CTRL-A>

Parameter: n 0 to $7F (0 to 255) specifies an ASCII character code.

Refer to: CSTATUS, USERS, STREAMCA, STREAMDB, LCSTREAM

STREAMSW selects the characters used by both the TNC and the user to show that a new connection channel is being addressed.

The character can be PASSed in the Converse Mode. This character is always ignored as a user-initiated channel switch in the Transparent Mode; it just flows through as data.

NOTES:

1. You cannot change the outgoing channel while the TNC is active or "on-line" in the Transparent Mode.
2. To switch channels, ESCAPE to the Command Mode. Then enter the Converse Mode to use the **STREAMSW** command.

Refer to **STREAMCA** for further use of **STREAMSW**.

STREAMCA ON/OFF

Command: STREAMCA ON/OFF
Mnemonic: STREAMC
Default: OFF

Parameter: ON Call sign of the other station is displayed in multiple connection operation.
OFF Call sign of the other station is not displayed.

Refer to: CSTATUS, USERS, STREAMSW, STREAMDB

STREAMCA displays the call sign on the "connected-to" station after the channel identifier.

Set STREAMCA ON if you intend to operate multiple connections (as opposed to having your "host" computer operated multiple connections).

STREAMCA is especially useful when you are operating with multiple connections. Using STREAMCA is similar to using MRPT to show digipeater paths when you are monitoring the channel.

EXAMPLES:

1) When STREAMCA is OFF, the monitored activity looks like this:

```
:A hi bob
hi ted how goes it?
:B *** CONNECTED to W6YEY
```

```
:B must be a dx record.
:A unreal ted
:B big band opening
```

2) When STREAMCA is ON, the same activity looks like the following.

```
:A: W6UOU: hi bob
hi ted how goes it?
:B: W6YEY:*** CONNECTED to W6YEY
:B: must be a dx record.
:A: unreal ted
:B: W6YEY:big band opening
```

With STREAMCA ON, ":A" becomes ":A:<call sign>:"

STREAMDB ON/OFF

Command: STREAMDB ON/OFF
Mnemonic: STREAMD
Default: OFF

Parameter: ON Received STREAMSW characters appear twice (doubled).
OFF Received STREAMSW characters appear once (not doubled).

Refer to: CSTATUS, USERS, STREAMSW, STREAMCA, LCSTREAM

STREAMDB displays received STREAMSW characters as double characters.

In the following example, STREAMDB is ON and STREAMSW is set to "|":

```
|| this is a test.
```

The sending station actually transmitted:

```
| this is a test.
```

The same frame received with STREAMDB OFF would be displayed as:

```
| this is a test.
```

✅ **NOTE:** Set STREAMDB ON when you operate with multiple connections so you can tell the deference between STREAMSW characters received from the other stations and STREAMSW characters generated by your TNC.

✅ **NOTE:** STREAMSW characters must not be one of the channel numbers (A through K) for this command to function properly.

TRANS

Command: TRANS
Mnemonic: T
Default: Immediate Command

Parameter: none

Refer to: COMMAND, CMDTIME, CONVERSE

TRANS is an immediate command that switches the TNC from the Command Mode to the Transparent Mode. The current state of the radio link is not affected.

Transparent mode is primarily useful for computer communications. In the Transparent Mode, "human interface" features such as input editing, echoing of input characters, and type-in flow control are disabled.

Use the Transparent Mode when you need to transfer binary or other non-text files.

TRACE ON/OFF

Command: TRACE ON/OFF
Mnemonic: TRAC
Default: OFF

Parameter: ON Trace function is enabled.
OFF Trace function is disabled.

The TRACE command activates the AX.25 protocol display. If TRACE is ON, all received frames are displayed in their entirety, including all header information.

✅ **WARNING:** Be careful when you use mnemonic - do not use "**TRA**"!
TRA cause the Controller to change to the Transparent Mode.

The TRACE display is shown as it appears on an 80-column display. The following monitored frame is a sample:

```
W2XYZ* > TESTER <UI>:

Byte            Hex                     Shifted ASCII
====            ===                     =============
000: AA88AA6A8 8AA460AE 6494AA0 406103F0 TESTERoW2XYZ

ASCII
=====
o.x  .....`d...@a..
```

The byte column shows the offset into the packet of the beginning byte of the line.

The hex display column shows the next 16 bytes of the packet, exactly as received, in standard hex format. The shifted ASCII column decodes the high-order seven bits of each bytes as an ASCII character code.

The ASCII column decodes the low-order seven bits of each bytes as an ASCII character code.

In standard AX.25 packet:

1. The call sign address field is displayed correctly in the ASCII column.
2. A text message is displayed correctly in the ASCII column.
3. Non-Printing characters and control characters are displayed in both ASCII fields as period (".").

You can examine the hex display field to see the contents of the SSID byte and the control bytes used by the protocol.

TRFLOW ON/FF

Command: TRFLOW ON/FF
Mnemonic: TRF
Default: OFF

Parameter: ON Software flow control for the computer or terminal can be activated in the Transparent Mode.
OFF Software flow control for the computer or terminal is disabled in the Transparent Mode.

Refer to: START, STOP, XFLOW, TXFLOW

If **TRFLOW** is ON, the type of flow control used in the Transparent Mode is determined by how **START** and **STOP** are set.

If **TRFLOW** is OFF, only "hardware" flow control (CTS, RTS) is available to the computer and all characters received by the TNC are transmitted as data.

If **START** and **STOP** are set to $00, the user Stop and User Start characters are disabled - hardware flow control must be used by the computer.

If **TRFLOW** is ON, and **START** and **STOP** are set to values other than zero, software flow control is enabled for the user's computer or terminal. The TNC responds to the User Start and User Stop characters (set by **START** and **STOP**) while remaining transparent to all other characters from the terminal.

Unless **TXFLOW** is also ON only hardware flow control is available to the TNC to control output from the terminal.

✅ **NOTE:** Even though the Pico Tnc does not have CTS, RTS wired to the TTL serial or USB serial ports, this feature would most likely not be used today unless hooked to a retro computer as most of todays computers are so fast they can easily accept any data the TNC provides at Full Speed.

TRIES n

Command:	TRIES n
Mnemonic:	TRI
Default:	0

Parameter: n 0 to 15 specifies the current RETRY level on the selected input channel.

Refer to: RETRY

TRIES retrieves (or forces) the count of "tries" on the data channel presently selected.

If you type tries without an argument, the TNC returns the current number of tries if an outstanding unacknowledged frame exists. If no outstanding unacknowledged frame exists, the TNC returns the number of tries required to get an ACK for the previous frame.

If **RETRY** is set to zero(0), the **TRIES** command always returns zero.

Use TRIES for gathering statistics on a given path or channel.**TRIES** is especially useful for computer-operated stations (such as automatic message-forwarding stations) using less-than-optimal, noise HF or satellite channels or paths.

Using **TRIES** under these conditions automatically optimizes the **PACLEN** and **MAXFRAME** parameters.

If you type **TRIES** with an argument, the 'tries' counter is forced to the entered value. Using this command to force a new count of tries is not recommended.

TXDELAY n

Command: TXDELAY n
Mnemonic: TX
Default : 50 (500ms)

Parameter : n 0 to 120 specifies 10 mS intervals.

Refer to: AXDELAY, AXHANG

The **TXDELAY** command tells the TNC how long to wait before sending packet frame data after keying your transmitter's PTT line.

All transmitters need some amount of start-up time to put a signal on the air; some need more, some need less.

Some general rules apply to these:

1. Crystal-controlled radios with the diode T/R switches are faster
2. Synthesizer radios need time for phase-lock-loop to lock.
3. Radios with mechanical transmit / receive relays need more time.
4. External amplifiers that use RF-driven relay switching usually require you to increase **TXDELAY** to allow for the additional delays.

Experiment to determine the best **TXDELAY** value for a specific radio.

TXDELAY can also compensate for certain characteristics of the radio used by the station which you are communicating with.

If the distant station's radio has slow AGC recovery or squelch release times when it is switching from transmit to receive, increasing your **TXDELAY** may reduce retries and improve throughput by retarding the start of your data until the other receiver has reached full sensitivity.

✅ **Warning:** Because of the way the TNC Emulation code works, it is important to keep this value at 50 or above! If not you may have problems with the tnc code hanging up.

✅ **NOTE:** The **TXDELAY** command has another special funtion. Whenever the TNC emulator detects a change to **TXDELAY** it saves all of the TNC Parameters to FLASH MEMORY. It is recommended to set it to 50 but if it is already at 50 you can set it to 51 or 49 to save your parameters to the pico flash memory. If you care to you can do it again and set it back to 50 afterwards and it will save again. ***Again to save your TNC parmeters change TXDELAY.***

TXFLOW ON/OFF

Command: TXFLOW ON/OFF
Mnemonic: TXF
Default: OFF

Parameter: ON Software flow control for the TNC can be activated in the Transparent Mode.
OFF Software control for the TNC is disabled in the Transparent Mode.

Refer to: XON, XOFF, XFLOW, TRFLOW

When **TXFLOW** is ON, the setting of **XFLOW** determines the type of flow control used in Transparent mode.

When **TXFLOW** is OFF, the TNC uses only hardware flow control; all data sent to the terminal remains fully transparent.

When **TXFLOW** and **XFLOW** are ON, the TNC uses the Start and Stop characters (set by XON and XOFF) to control the input from the terminal.

Unless **TRFLOW** is also ON, only hardware flow control is available to the computer or terminal to control output from the TNC.

UNPROTO call1[VIA call2[,call3...,call9]]

Command: UNPROTO call1[VIA call2[,call3...,call9]]
Mnemonic: U
Default: CQ

Parameter : call1 Call sign to be placed in the TO address field. Call2-9 Optional digipeater call list, up to eight calls.

Refer to: **CONNECT**

UNPROTO sets the digipeat and destination address fields or packets sent in the unconnected (unprotocol) mode.

Unconnected packets are sent as un-sequenced I-frame with the destination and digipeat fields taken from "call1" through "call9" options. When a destination is not specified, unconnected packets are sent to "CQ".

You can monitor unconnected packets sent from other packet stations by setting MONITOR ON.

USERS n

Command: USERS n
Mnemonic: US
Default: 1

Parameter: n 0 to 10 specifies the number of active simultaneous connections that can be established with your TNC.

Refer to: **STREAMSW**

USERS only affects the way that incoming connect requests are handled. It does not affect the number of connections you initiate with your TNC. For example:

USERS 0 Allows incoming connections on any free logical channel.

USERS 1 Allows incoming connections on logical channel 0 only.

USERS 2 Allows incoming connections on logical channel 0 and 1.

USERS 3 Allows incoming connections on logical channel 0, 1 and 2, and so on, through USERS 10.

WRITE

Command: WRITE
Mnemonic: W
Default: Immediate Command

Parameter: none

Refer to: "Using the Bulletin Board Feature" in the Operating Instruction Manual.

WRITE command posts a message on the BBS.

XFLOW ON/OF

Command: XFLOW ON/OFF
Mnemonic: X
Default: ON

Parameter: ON XON/XOFF (software) flow control is activated.
OFF XON/XOFF flow control is disabled - hardware control is enabled.

Refer to: TRFLOW, TXFLOW, START, STOP, XON, XOFF, FLOVER, RSFLOW

When XFLOW is ON, software flow control is in effect; it is assumed that the computer or terminal will respond to the TNC's start and stop characters defined by the XON and XOFF commands.

When XFLOW is OFF, the TNC uses hardware flow control command on the CTS and RTS line.

✅ **NOTE:** Hardware flow control is not available on the Pico Tnc do to limitations of the TTL serial port.

XMITOK ON/OFF

Command: XMITOK ON/OFF
Mnemonic: XMITO
Default: OFF

Parameter: ON Transmit functions (PTT line) are active.
OFF Transmit functions (PTT line) are disabled.

Refer to: CONOK, DIGIPEAT, MYCALL

When **XMITOK** is OFF, the PTT line to your transmitter is disabled; the transmit function is inhibited. All other TNC functions remain the same. Your TNC generates and sends packets as requested, but does not key the radio's PTT line.

Use the **XMITOK** command at any time to ensure that your TNC does not transmit.

Set **XMITOK** OFF if you are absent and wish to leave your TNC on as a channel activity monitor.

Set **XMITOK** OFF for testing in loopback or direct wire connections when PTT operation is not required.

XOFF n

Command: XOFF n
Mnemonic: XO
Default: $13 <CTRL-S>

Parameter: n 0 to $7F (0 to 127 decimal) specifies and ASCII code.

Refer to: START, STOP, XFLOW, XON, TXFLOW

Use **XOFF** to select the stop character to be used to stop input from the computer or terminal.

✅ **NOTE:** You can entire the code in either hex or decimal.

The Stop character default value is <CTRL-S> for computer data transfers.

If you are operating in the Converse Mode and there is a chance that activity might fill the TNC's buffers, you can define the Stop character as <CTRL-G> ($07), which "beeps" many terminals.

XON n

Command: XON n
Mnemonic: XON
Default: $11 <CTRL-Q>

Parameter: n 0 to $7F (0 through 127 decimal) specifies an ASCII character code.

Refer to: START, STOP, XFLOW, XOFF, TXFLOW

XON selects the TNC start character that is sent to the computer or terminal to restart input from the device.

✅ **NOTE:** You can entire the code in either hex or decimal.

The Start character default value is <CTRL-Q> for computer data transfers.

Glossary

AFSK (Audio Frequency Shift Keying) — Modulation method using two tones (mark and space) to represent digital data; typically 1200 Hz and 2200 Hz for Bell 202 packet.

AX.25 — Amateur Packet Radio link-layer protocol based on X.25, handling framing, addressing, and error checking.

AX.25 Level 2 — The full implementation of the AX.25 link layer with retransmission and acknowledgment logic.

Baud Rate — Symbol rate of data transmission (usually 1200 baud on VHF packet).

Beacon / BTEXT — Automatic broadcast message periodically sent by a station to announce presence or status.

Bell 202 — The 1200 baud AFSK standard using 1200 Hz (mark) and 2200 Hz (space) tones.

BBS (Bulletin Board System) — Packet-based message storage and retrieval system accessible by radio.

Callsign / MYCALL — Licensed station identifier embedded in every packet transmission.

Checksum / CRC — Mathematical value used to detect transmission errors in packet frames.

Command Mode — The TNC mode where user commands are entered to control or configure the unit.

Converse Mode — Operational mode where typed text or data is transmitted as packets during a live connection.

CSMA (Carrier Sense Multiple Access) — Technique in which the TNC monitors the channel and transmits only when clear.

DCD (Data Carrier Detect) — Indicates that the TNC is detecting a valid signal or carrier on the channel.

Digipeater — A node that automatically repeats packets to extend range or reach other stations.

DWAIT — The waiting period between detecting a clear channel and starting transmission.

FRACK (Frame Acknowledgment Time) — Time the TNC waits for an acknowledgment before retransmitting a frame.

Full Duplex — Simultaneous transmit and receive operation using separate frequencies.

HF / VHF Packet — Packet operation on high- or very-high-frequency bands (300 baud HF, 1200 baud VHF).

KISS (Keep It Simple, Stupid) — Minimal framing protocol that lets a computer handle AX.25 logic while the TNC only modulates/demodulates.

MBOD — Command to enable or disable the TNC's built-in PBBS (mailbox).

Monitor / MONITOR ON — Displays all received packets for observation or debugging.

PBBS (Personal Bulletin Board System) — Simplified mailbox built into a TNC for message exchange without an external computer.

Persist / PERSIST Command — Defines transmission probability in p-persistent CSMA; works with SLOTTIME.

PTT (Push-To-Talk) — Control line that activates the radio's transmitter when sending.

Retry / RETRY Command — Number of times the TNC resends an unacknowledged frame before giving up.

SLOTTIME — Interval length used with PERSIST to determine channel access timing.

SSID (Secondary Station Identifier) — Numeric suffix appended to a callsign (e.g., N0CALL-1) to identify multiple services or devices.

TNC (Terminal Node Controller) — Device that converts digital data into radio tones and back, implementing AX.25.

Transparent Mode — Pass-through mode where binary data is sent without interpretation by the TNC.

TXDELAY — Adjustable delay between keying the transmitter (PTT) and sending data, ensuring the transmitter reaches full power.

UI Frame (Unnumbered Information Frame) — AX.25 frame type for unconnected transmissions (e.g., beacons, APRS).

UNPROTO Command — Defines the destination path for UI frames (used for beacons or APRS).

Z80 Emulation — Simulation of the classic Z80 CPU to run legacy TNC2 firmware on the Pico TNC.

Acronym Reference (Alphabetical)

Acronym	Meaning
AFSK	Audio Frequency Shift Keying
APRS	Automatic Packet Reporting System
AX.25	Amateur X.25 Protocol
BBS	Bulletin Board System
BTEXT	Beacon Text
CRC	Cyclic Redundancy Check
CSMA	Carrier Sense Multiple Access
DCD	Data Carrier Detect
HF	High Frequency
KISS	Minimal Framing Protocol
MBOD	Mailbox On/Off
PBBS	Personal Bulletin Board System
PTT	Push-To-Talk
RX	Receive
SSID	Secondary Station Identifier
TNC	Terminal Node Controller
TX	Transmit
UI	Unnumbered Information (Frame)
VHF	Very High Frequency
Z80	8-bit Microprocessor by Zilog

Command Index

Pico TNC Assembly Guide

The Pico TNC can be built on a breadboard or perfboard, but the easiest method is to use the **printed circuit board (PCB) design** included in the github repository. A pre-designed PCB removes guesswork, makes wiring reliable, and results in a compact, durable unit. Also included in the repository are 3d .stl files for a case that can be printed on a 3d printer.

You have three options for getting a PCB:

1. **Use the provided Gerber files** — send them to a PCB manufacturer of your choice.
2. **Order from OSH Park** — a shared project is published there for quick ordering and can be found by searching on their site for **PICOTNCEMU.** You can also view and order the pcb by going to this URL: **https://oshpark.com/shared_projects/5Q3zsFlU**
3. **Fabricate locally** — some hackerspaces and clubs have PCB milling or group orders.

Bill of Materials (BOM)

Below is the list of components typically required. Always check the latest schematic (`schematic.png`) in the GitHub repo in case of updates.

Quantity	Part	Notes
1	Raspberry Pi Pico with Male Header	Main processor
1	PCB (from Gerber files or OSH Park)	Pico TNC board
1	3.5mm **PJ320D** Horizontal TRRS Jack	TX/RX Audio, PTT to radio
1	3.5mm **PJ-307** TRS jack (optional)	TTL Serial Port
1	USB connector (Pico provides)	For PC connection & power
2	Protection diodes 1N4148 or similar	For audio input
2	Resistors 10k 1/8 watt	Biasing audio input
2	Resistors 4.7k 1/8 watt	PTT Bias
1	Resistor 22K or 4.7K (OPTIONAL HT)	For Keying HT
1	Resistor 100 ohm 1/8 watt	For pwm output filter
1	10k Potentiometer	Tx Audio Adjust
3	Capacitors 0.1 µF	Filtering
4	LEDs 3mm 4 different colors	Status indicators
4	Resistors for LEDs 1kΩ 1/8 Watt	Sets LED brightness
1	4 Pin DIP Switch Piano Configuration	Options Switches
1	2N3904 NPN Transistor TO-94 Package	Keys Radio
Headers	Female pin headers	For mounting the Pico
Misc.	Enclosure (optional)	Protects the unit

(Confirm exact values from schematic before ordering — the above is a generic outline.)

Step-by-Step Assembly

Follow this order to make soldering and troubleshooting easier:

1. **Inspect the PCB**
    - Check the board for defects, scratches, or broken traces.
    - Familiarize yourself with component labels (TRRS jack, Pico header, resistors).
2. **Solder the Audio Jacks first.**
    - Apply solder to the surface mount pads of the 3.5mm TRRS audio jack. Then carefully remove as much solder as possible using solder braid so as to leave them flat and tinned.
    - It is recommended to sparingly put a small drop of super glue in the center of where the surface mount Jack will sit avoiding the solder pads before placing the Jack. Then apply heat to the pins and a small amount of solder to solder each pin of the Jack.
    - Insert and solder the remaining 3.5mm TRS Jack.
3. **Solder the small passive parts first**
    - Install **resistors** and **diodes**. If 1/8 watt resistors are not available ¼ watt can be used but they will need to be inserted in a radial fashion (stand up fashion).
    - Then install the 3 small.1uf **capacitors.**
    - Insert and Solder the DIP Switch. Make sure that switches are facing the edge of the PC Board.
    - This keeps the board flat while soldering.
4. **Install pin headers for the Pico**
    - Insert femmale headers into the PCB, place the Pico onto the headers to keep them straight and oriented properly, and solder from underneath.
    - If soldering the Pico directly to the board double-check orientation: USB port should face opposite side of the Dip Switch.
5. **Mount Leds**
    - Take a single LED and bend the pins uniformly so that they are bent about 1mm from the body at 90 degrees. Make sure the short pin is going into the square hole of the pcb pad for the led, that is Ground. Solder only 1 pin and test for fit in the case if you are using a case. You may have to adjust the bend point for your case.

- Once you have determined the Bend point of the LED for your case solder the remaining leg and bend the rest of the Led's in the same fashion and carefully solder into their positions. Test for fit as you go.
- Add any extra headers or serial connectors if your board supports them..

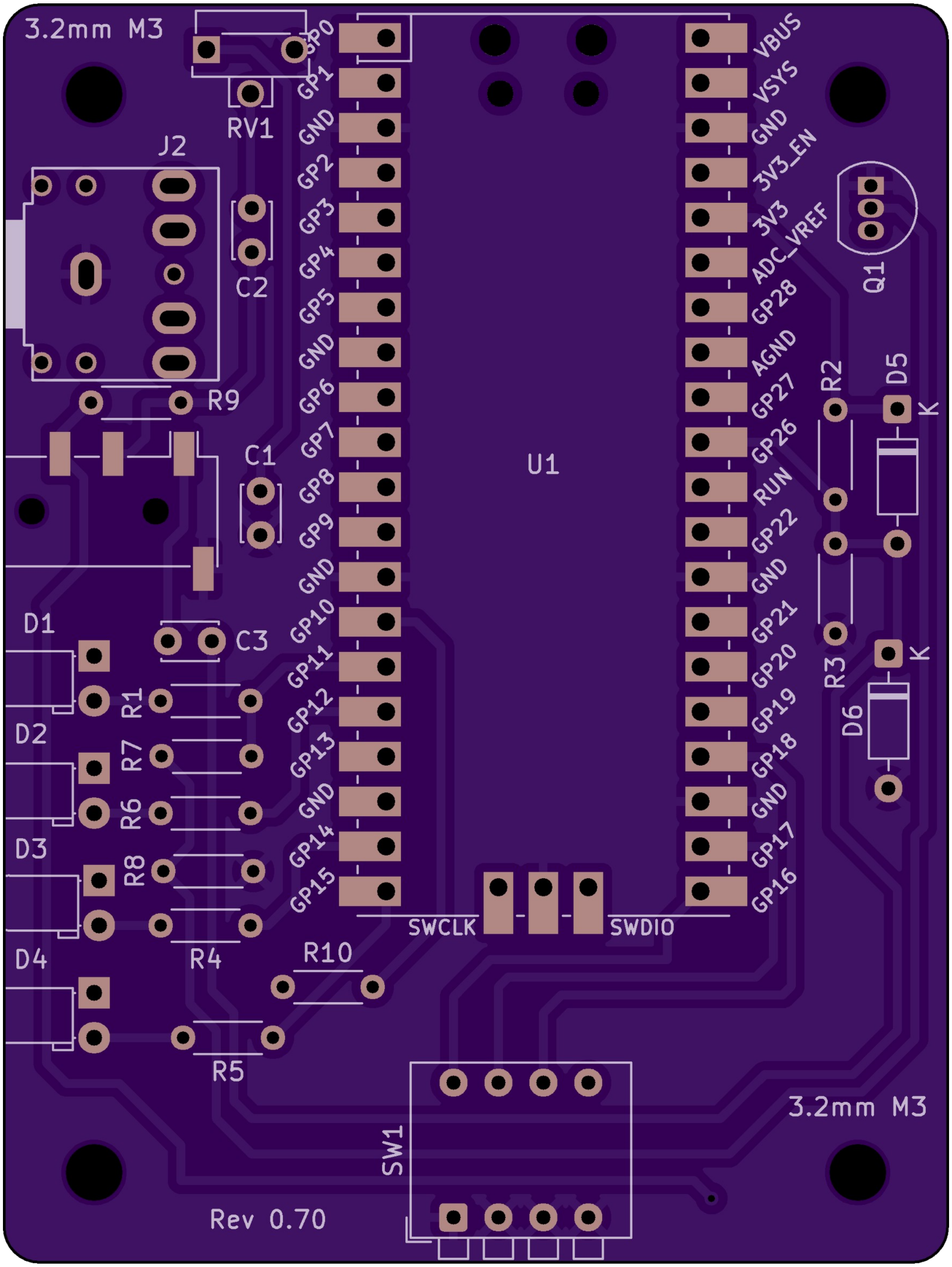
6. **Final checks**
   - Inspect all solder joints with good light.
   - Make sure there are no bridges (shorts) between pads.
   - Plug in the Pico (without the radio connected yet) and verify that it powers up.
7. **Connect to radio**
   - Attach TRRS cable to your radio's mic/speaker/PTT as described in the "Connecting to a Radio" section of this manual.
   - Start with low audio levels and test with a dummy load.

Tips for Beginners

- **Use flux and a clean iron tip** — it makes soldering much easier.
- **Work from low-profile to tall parts** — resistors → IC headers → connectors.
- **Check orientation carefully** — diodes and LEDs must face the correct way.
- **Don't overheat the Pico** — solder the headers quickly to avoid lifting pads.
- If in doubt, compare your build against the schematic in the repo.

✅ With just a handful of parts, you'll have a fully functional Pico TNC ready to flash and get on the air.

PC Board


3.2mm M3
RV1
J2
C2
R9
C1
U1
D1
C3
R1
D2
R7
R6
D3
R8
D4
R4
R10
R5
SW1
Rev 0.70
3.2mm M3
SWCLK
SWDIO
GP0
GP1
GND
GP2
GP3
GP4
GP5
GND
GP6
GP7
GP8
GP9
GND
GP10
GP11
GP12
GP13
GND
GP14
GP15
VBUS
VSYS
GND
3V3_EN
3V3
ADC_VREF
GP28
AGND
GP27
GP26
RUN
GP22
GND
GP21
GP20
GP19
GP18
GND
GP17
GP16
Q1
R2
D5
K
R3
K
D6

🔧 Pico TNC Assembly Checklist (PCB Version)

Follow the numbered order for easiest assembly. Refer to the PCB silkscreen or schematic for part locations.

Step 1 — Smallest Parts First

1. **Audio Jacks – S**older the surface mount 3.5mm TRRS Jack (J1) first. Then the 3.5mm TRS TTL Jack (J2).
2. **Resistors** – Insert and solder all resistors (low profile, usually labeled R1, R2, etc.).
3. **Diodes** – Install protection diodes (mind polarity: stripe = cathode).
4. **Small capacitors** – Add ceramic or electrolytic caps (if any).

✅ *Tip: Check resistor color codes with a meter before soldering.*

✅ *Tip: Pre tin the PCB solder pads and wipe with solder wick where the 3.5mm (J1) Jack mounts.*

✅ *Tip: Place a small drop of super glue between the pcb and the 3.5mm (J1) Jack to hold it in place and add extra strength. Place carefully and minimally to avoid the solder pads.*

✅ *Tip: Check resistor color codes with a meter before soldering. 1/8 watt resistors install flat in axial fashion, you can use 1/4 watt also but they must be installed vertically in radial fashion.*

Step 2 — IC Headers / Pico Mount / Dip Switch

4. **Pin headers for Raspberry Pi Pico** – Insert into PCB, place Pico on top, then solder from the underside.
    - Ensure correct orientation (USB facing board edge).
    - Keep the Pico removable if you prefer (use female headers).
    - Make sure Pico usb port faces away from Dip Switches if soldering to board.
    - Make sure Dip Switch pins face edge of pc board.

Step 4 — Other Components

8. **LED indicator** – Insert LED, long leg = +, short leg = ground.
9. **Optional Resistor R9 –** Several radio and cable combinations have been confirmed to work without this resistor installed. Suggest to test TNC & Radio combo before installing R9. R9 is usually only installed when the rig does not have a PTT but uses a resistance change in the microphone input to detect PTT. This is usually the case with some HT's and other rigs.

✅ *Tip: Double check the Polarity of LEDS before bending the LED Legs.*

✅ *Tip: The short leg on the LED (Ground Leg) goes into the square pcb hole.*

Step 5 — Final Assembly

10. **Inspect solder joints** – Look for cold joints or bridges.
11. **Power test** – Plug in via USB, ensure Pico powers up.
12. **Radio test** – Connect audio/PTT cable to radio, start with low levels, and verify transmit/receive works.

Revision History

- Version 1.0: September 29, 2025 – Initial draft based on project documentation from GitHub repository.